# The Particle Zoo under Declared Matter Premises: One Generation, Interaction Channels and Mass Diagnostics

B. Müller Mario Poneder Alexander Osika Kai Xue Daniel Ari Friedman

October 26, 2026

**Paper release:** r2054 **Released:** October 26, 2026

**Author affiliations:** Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks; Daniel Ari Friedman, Active Inference Institute and FractiAI.

**Abstract**

A particle spectrum requires more than a gauge representation: it requires physical fields, their dynamics, a vacuum and a mass observable. We separate these requirements for finite Observer Patch Holography constructions. A specified twelve-current response, with a declared spin lift and exterior module, gives the fifteen internal states of one Standard Model generation and its three charged-fermion Higgs channels. Their family coefficients are free. Calibrated finite actions permit observer-record recovery of supplied mass parameters; the tested uncalibrated repair prescriptions fail the charged hierarchies. We give exact normalization bounds and restricted nonidentifiability results explaining those failures. Conditional flavor, electroweak, clock and pole constructions identify the additional hypotheses needed for physical masses. We also distinguish finite record continuation from particle propagation and state the continuum, kinetic-normalization and source-selection requirements. The construction establishes conditional representation and readout results, rather than a source-derived measured particle spectrum or a complete interacting Standard Model.

**Interpretation and state variables.** OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

**Basis, premise labels and paper set.** OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Because nothing lies outside that structure, recovered structure needs no a-priori justification, nothing unrecovered may be assumed, and every change of description is implemented by the structure's own operations. Agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with its endogeneity clause; and least-informative state selection. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

**Time conventions.** Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

**The executable completion hypothesis.** The concrete candidate is one simple, fully specified local repair and memory law on the proposed carrier architecture whose continuing observers admit the observed interacting chiral $3+1$-dimensional field description and a compatible coarse-scale Einstein description. One process must supply the records, clocks, matter, energy and responsive geometry together. This is a stronger executable hypothesis within OPH's closure proposal, not a theorem that the present C/E implementation or every admitted law realizes our world.

The preferred matter mechanism is localized topological defects of native overlap transport. Stable massive matter is the first target; unstable species require metastable or internal excitations and derived decays, and massless modes require their own same-process interpretation. The defect's excess maintenance dynamics must give the same rest energy, kinetic normalization and stress measured in propagation, interactions and gravity. Repair counts alone are not energy. A massless photon makes the full infinite-volume theory gapless; massive thresholds and any pure Yang–Mills gap require their own sectors and limits. Confinement precludes treating gluons as observable free massless particles. A possible quantum graviton is outside Standard Model content and is not proved by the classical theorem.

The observer's algebra and faithful reference define dimensionless modular flow. The candidate must implement it, derive compatible local clock/energy maps and realize the Einstein construction's geometric modular, conserved stress, entropy/area and fixed-proper-volume stationarity premises on that same refinement. Stationary repair or record entropy alone does not provide that geometric variation. Exact limiting observable laws are the target; finite spectrum and interaction comparisons use declared matching and physical uncertainty.

**Foundations and the next joins.** The existing native rank-three slow-response geometry, candidate modular clock, common screen-field generator and conditional smooth $1+3$ scalar description

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/docs/BASIS.md

are retained foundations. The field extension uses declared radial and encoded-field/control choices; rank three of the local band does not alone derive three spatial field dimensions. Finite interacting histories and the conditional Einstein tensor kernel are further reusable components. Physical calibration, the common defect/energy identification, chiral interacting refinement and responsive geometry remain to be constructed.

The completion program orders these joins and assigns each required finite theorem a model-premise proof obligation. Irrelevant or redundant theorem branches leave the active chain with their provenance preserved. Its first gate freezes the admissible program class; independently checked universal obstructions or unavoidable empirical contradictions can stop that class. Candidate failures, open maps and resource stops keep their narrower meanings. The hypothesis, dependency DAG, retained foundations and issue owners are recorded in the public repository's `docs/OPH_EXECUTABLE_HYPOTHESIS.md` and `docs/COMPLETION_PROGRAM.md`.

# 1 Particle structure and physical mass

Observer Patch Holography (OPH) models observations through finite accessible algebras, retained records and local repair. The regulated observer construction provides explicit source operators and instruments [11, 12]. This paper asks which particle statements follow from those operators and which require an additional field-theory attachment. The gauge classification is developed in Ref. [10]; the finite matter decoder is restated below so that its hypotheses and its limitations can be assessed without an inferred identification of internal states with laboratory particles.

The principal conclusions are:

1. A specified closed twelve-current law admits a conditional Standard Model matter and Higgs representation. Its symmetry identities permit arbitrary family Yukawa matrices.

2. Calibrated actions realize charged-matter structure and permit recovery of their supplied mass parameters from simulated observer records. This is a realization and readout test, not a derivation of the calibration inputs.

3. The uncalibrated response laws tested here do not reproduce the measured charged spectrum. Exact exclusions apply to stated prescriptions, including normalized projector averages and an affine family of quadratic responses.

4. The physical chiral interacting continuum, source-selected vacuum, kinetic forms, Yukawa coefficients and physical mass/clock comparison are additional requirements. No measured particle spectrum is derived.

A finite internal eigenvalue may be a representation label, a memory level, a relaxation rate, a Hamiltonian coefficient or a mass. Those meanings require different observation maps. In particular, a repair iteration counter is not an operational clock, and a finite relaxation gap is not a relativistic mass-shell pole. The word *source-derived* below requires the complete source program to determine an object independently of its empirical target, including the ancestry of its discrete choices and coefficients. Absence of a target file at runtime does not establish this independence.

A particle identification must begin with one excitation or sector of one continuing source, and preserve that object through its charge, propagation, detector, kinetic and interaction maps. The different representation labels, memory levels, decay rates and benchmark field poles below cannot be pooled into a spectrum by choosing the closest numerical row. For a stable excitation in an appropriate vacuum regime, nonzero response residue and dispersion must accompany the quantum

numbers; unstable, confined and infrared charged sectors require their own observable definitions. A topological-defect interpretation additionally needs an actual invariant of the admitted native updates. Nonzero repair frustration or resource expenditure alone does not identify inertial mass. The source must connect the same excitation's excess energy and clocked propagation, including kinetic normalization. The current construction contains no established native defect-to-physical-mass map.

The preferred working hypothesis is a localized, persistent topological defect of native overlap transport whose continued maintenance contributes the excess rest energy of that same excitation. Stable massive matter is the first target; unstable species would require defect excitations and derived decay channels. A boundary invariant, localized dynamics, spin/statistics, currents and the common energy/clock relation must be constructed before testing mass ratios. This hypothesis motivates the native decision experiment below; it is not a consequence of mass–energy equivalence alone.

## 2 Source and attachment premises

**A1 (oriented observer-patch federation and spherical support).** There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator $r$ in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve $N_r$; a finite edge-midpoint refinement support $S_r$ realized orientation-preservingly in $S^2$ with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

**A2 (observer agreement).** Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and

$\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart $o$, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism $a$, A2 supplies a closed path $\gamma_a$ and one projective implementer $[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

**A3 (conditional maximum randomness).** Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator $r$ a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family $\tau_r$, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise reference.

**The closure hypothesis.** The theorem premises A1–A3 state the library's hypotheses on the carrier and its response, on internal transport and on local state selection. The closure hypothesis states that the Universe is a timeless, self-referential structure whose observers recover the specification of their own world from their records and realize it, so that the description and the described system are one object. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The closed object is the complete inhabited history, whose internal events and responses to interventions are its proposed intrinsic computation. This reading requires no external computer or playback time. It is distinct from fixedness of a repair state, a public record, or a probability law, and does not itself construct the history or prove a final cosmic endpoint.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point per declared map on its analytic domain. Interpreting that root as the laboratory coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at $3.0 \times 10^{-4}$ relative for the displayed map and at $2.5 \times 10^{-6}$ for the gauge-width variant, as diagnostics of the declared source maps. The latter absolute residual is $3.3904 \times 10^{-4}$, compared with experimental standard uncertainty $2.1 \times 10^{-8}$; this is not agreement at measured precision. The gauge width supplies 99.2 percent of the gap between the bare root and the measured value; the residual $3.3904 \times 10^{-4}$ is addressed to same-scheme hadronic transport to the Thomson endpoint.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct $N$, and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the A1–A3 source contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\left(\frac{6\pi}{P\,\alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival

factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\mathrm{pres}} = N_0 \left(1 - \frac{P}{24}\right), \qquad \ln \frac{N_{\mathrm{pres}}}{\pi} = \frac{6\pi}{P\,\alpha_U(P)} + \ln\left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\mathrm{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is $3.2921 \times 10^{122}$ and the exponential candidate is $3.3001 \times 10^{122}$. The full weighted Planck base-$\Lambda$CDM chain gives the comparison coordinate $3.3129 \times 10^{122}$. The respective residuals are $-0.63$ percent and $-0.39$ percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

These premises organize the conditional OPH mathematical framework. They do not select every quantitative particle completion. The screen-first spherical support, the local twelve-port carrier and the federation overlap nerve are distinct objects; identifying their physical carriers consumes the separate incidence, access, overlap and refinement hypotheses [6, 10].

| Object | Hypothesis needed beyond its finite algebraic calculation |
|---|---|
| Gauge currents | Selection of the complete physically available response, rather than a chosen control subspace. |
| Matter module | Spin cover, exterior-sector choice, physical spin and CAR statistics, and a field interpretation. |
| Three families | Attachment of a protected rank-three band to three physical generations, with specified symmetry breaking. |
| Mass coefficients | A source-selected Higgs response distinguished from the kinetic metric, a vacuum and a common normalization. |
| Physical spectrum | Controlled chiral continuum, interactions, renormalization, infrared treatment, observable singularities and an operational clock. |

The quantitative pixel branch uses a dimensionless closure coordinate $P$. An interval contraction certificate establishes a unique root on the declared numerical domain; it does not identify that root with a physical endpoint. The distinction is essential: a scalar fixes a mass scale only after a source law maps it into a physical generator and a clock. The conditional transmutation formula below does not determine its running-coupling function $\alpha_U(P)$, cutoff or matching prescription. Their field content, normalization and empirical inputs must be supplied separately; they cannot be inferred from the finite gauge algebra. The comparison branches and their limitations are stated in Section 6.3.

The universe-level capacity relation is

$$N = \log M_0(\mathfrak{U}_N), \qquad \mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N = \log D_\star.$$

Here $M_0$ is the multiplicative size of a correctable public record code. The finite readback is a compound-confusability independence number under a declared coupled checkpoint family [4]. Deflation, monotonicity and greatest-fixed-point iteration require their respective scalarization, faithful-carrier and confusability-reflection hypotheses. A fixed-cutoff capacity calculation does not supply the universe-level carrier or the common screen/electroweak load identification.

# 3 The current response and its matter representation

The native protected geometric object has type $3 \oplus 3'$. Twelve ports count interfaces; they do not count independent Lie generators. The tested native exchange response closes at seventeen currents. A specified law instead closes at

$$\mathfrak{u}(3) \oplus \mathfrak{so}(3), \qquad \dim = 12.$$

An additional coherent Hamiltonian filter has an exact invertible control-coordinate map into this same algebra. This is a conditional control construction, not a derivation that ordinary C/E state repair selects the physical gauge response. The full exchange-plus-class-sum controller has eighteen directions. Its finite filters are invertible; a fixed-bias long-time limit has six observable directions, whereas a joint bias/update limit with bounded drive approaches twelve. The limit and supplied baths must be specified before making a physical current claim.

The finite formal results are implications of their typed premises. An injective twelve-port response map has dimension twelve; it does not show that the complete native response admits that map. The Lie-type selection also consumes compact-simple classification data, a bound on the centre, and an exclusion of the zero-centre alternative. The committed matter instance supplies its spin lift, central weights and exterior mask. The family theorem compares a finite band table within its declared window. Checking these implications does not independently select their premises. In particular, observability of a relative central phase is not by itself a proof that the carrier action fixes the whole centre of the same complete response algebra.

## 3.1 Carrier covariance and physical rotations

The specified current construction uses the port vectors in its colour 3, while the family construction uses an equivalent carrier triplet on another factor. The real part of the colour generators implements rotations about those port axes. The carrier recharting premise acts by inner automorphisms on the current algebra. These facts establish carrier covariance; they do not establish which transformations observers recover as physical spatial rotations. Equivalent finite-group representations on different factors need not be one physical action. Conversely, declaring the same action to be both physical frame rotation and colour transport does not establish their required independence.

The existing spin–matter join tests a locked transport and then separates the factors with an explicitly chosen inverse internal transport. This is an additional construction choice. A physical identification must exhibit the spatial action, the internal action and their commutation on the same record/field domain, or calculate the observable consequences of a residual lock. The distinct $3'$ character alone does not settle the role of the colour 3. Likewise, a rank-three local carrier frame is not a proof of three-dimensional physical propagation on a federation. Neither a physical symmetry violation nor its resolution follows solely from the finite representation match.

## 3.2 Matter lift and interaction channels

The global finite gauge identification is $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$ under its stated faithful representation and central-kernel hypotheses. It fixes allowed representations, not their excitation energies. The following exact lift and interaction calculation connects the chosen current operators to the matter channels rather than relying on an abstract dimension match.

# 4 Matter and interaction readout from the finite response

The finite response determines more than the abstract gauge Lie type. Its matrix generators support an exact, conditional matter decoder and local interaction tensors. The distinction between a representation and a physical particle is essential: a particle identification also consumes its spin, statistics, propagator, interaction and observer-clock interpretation.

## 4.1 From response operators to a fifteen-state matter module

Let a response generator on the six-dimensional carrier have blocks $D = \mathrm{diag}(A, B)$, with $A \in \mathfrak{u}(3)$ and $B \in \mathfrak{so}(3)$. Write $B = \widehat{w}$. The declared spin lift and central trace redistribution give

$$L(D) = \mathrm{diag}\left(A, -\frac{i}{2} w \cdot \sigma - \frac{\mathrm{tr}\, A}{2} I_2\right).$$

This is a Lie-algebra homomorphism. The complete twelve-direction response and all sixty-six bracket pairs are checked exactly. The central direction, after clearing denominators and choosing the weak-positive orientation, has primitive integer character

$$q = \mathrm{diag}(-2, -2, -2, 3, 3).$$

The lift and trace redistribution are hypotheses. A native weak rotation through $2\pi$ is the identity on its vector carrier and acts as minus the identity on the lifted doublet. Thus the lifted module is not a subrepresentation of the original six-dimensional response. Global descent and a physical current interpretation require the corresponding additional identifications.

**Proposition 4.1** (Conditional matter decoder)**.** *For the lifted five-dimensional one-body action, the representation*

$$M = \Lambda^2 \mathbb{C}^5 \oplus \Lambda^4 \mathbb{C}^5$$

*has dimension fifteen. Its charge and quadratic Casimir operators identify the following irreducible components without a field-name input table.*

| Component | Color | Weak dimension | $Y = q/6$ | Dimension |
|---|---|---|---|---|
| $Q$ | 3 | 2 | 1/6 | 6 |
| $u^c$ | $\overline{3}$ | 1 | $-2/3$ | 3 |
| $d^c$ | $\overline{3}$ | 1 | 1/3 | 3 |
| $L$ | 1 | 2 | $-1/2$ | 2 |
| $e^c$ | 1 | 1 | 1 | 1 |

*Proof.* Split $\mathbb{C}^5 = V_c \oplus V_w$, of dimensions three and two and charges $-2$ and 3. Then

$$\Lambda^2(V_c \oplus V_w) = \Lambda^2 V_c \oplus (V_c \otimes V_w) \oplus \Lambda^2 V_w,$$

with integer charges $-4, 1, 6$. The degree-four summands are $\Lambda^3 V_c \otimes V_w$ and $\Lambda^2 V_c \otimes \Lambda^2 V_w$, of charges $-3, 2$. Color and weak Casimirs separate the representations, and Cartan weights distinguish the conjugate color character. Exact projectors give the same answer after a rational change of basis mixing unlike charge sectors. □

The auxiliary exterior construction has two additional invariant lines, the vacuum and top exterior degree. Removing them is part of the declared fifteen-state module. Charge traces alone do not remove neutral invariant states. Its even auxiliary parity also does not establish physical fermion statistics. Physical two-component Weyl fields require an independent CAR algebra and spin interpretation. The anomaly-free charge table and common central kernel are conditional representation results; they do not identify laboratory species or select a generation count.

### 4.2 One Higgs doublet and the exact interaction tensors

For two same-chirality physical Weyl fields, the antisymmetric spinor contraction makes the internal coefficient symmetric. The Lorentz-scalar internal pair space is consequently $\mathrm{Sym}^2 M$, of dimension 120. A color-Casimir kernel and weak-Casimir projector identify one doublet of hypercharge $-1/2$ and two of hypercharge $+1/2$. Their constituent integer-charge pairs are $(-4,1)$, $(-3,6)$, and $(1,2)$. These are interpolating channels; their existence supplies no bound-state pole, condensate or elementary-boson commutation relation.

**Proposition 4.2** (Conditional one-doublet interaction)**.** *Declare one scalar doublet $H$ of hypercharge $1/2$. Each of the three channels above has a one-dimensional invariant contraction space. They give*

$$Qu^cH, \qquad Qd^cH^*, \qquad Le^cH^*.$$

*The same scalar doublet and its conjugate occur in all three expressions.*

*Proof.* Restrict the exact pair generators to each color-singlet doublet. The tensor sum of that doublet generator and the generator of $H$, or its conjugate, has a one-dimensional common kernel. Equivalently, its coefficient matrices $C_\alpha$ satisfy, for each of the twelve source currents,

$$T^tC_\alpha + C_\alpha T + \sum_\beta (T_H)_{\beta\alpha} C_\beta = 0.$$

Here the conjugate action is $-T_H^t$ in the skew-Hermitian convention. The physical pairing kernel $\epsilon \otimes C_\alpha$ is antisymmetric. The identity $S^t\epsilon S = \epsilon$ for $S \in \mathrm{SL}(2,\mathbb{C})$ establishes its Lorentz scalar character. The executable checks the rotation and boost generators as well as all current identities. □

For three declared families the coefficients are three complex $3 \times 3$ matrices, with 27 complex entries before flavor-basis redefinitions. The pairing convention is

$$\mathcal{L}_{\mathrm{pair}} = \frac{1}{2}\epsilon_{AB}\psi_A^t K(H)\psi_B + \mathrm{h.c.}$$

up to an overall sign. With canonical kinetic normalization and $H = (0, v)$, its singular values are tree-level mass parameters; $v = v_{\mathrm{SM}}/\sqrt{2}$ in the conventional electroweak normalization. Exact matrix-unit tests cover arbitrary complex family coefficients. Their values, the Higgs potential and its vacuum are not selected by these identities. The optional invariant $(LH)(LH)/\Lambda$ has six complex symmetric family coefficients and gives $M_\nu = \kappa v^2/\Lambda$. It is an additional dimension-five effective interaction, with no selected coefficient, scale or ultraviolet completion.

If both family factors are identified with the same unbroken irreducible rank-three source representation and the Higgs is family-neutral, the exact commutant has dimension one. In that branch each Yukawa intertwiner is $yI_3$, giving equal singular values. A nonscalar background can split the band; choosing that background is an additional physical hypothesis. The source family-band theorem therefore does not imply the observed hierarchy.

### 4.3 Evidence and mass interpretation

A declared coherent filter on the native six-dimensional response admits an exact invertible twelve-control map into the same compact algebra; the spin/exterior and Higgs tensor maps above apply after this change of controls. It must not be identified with state cooling. For a rank-three projector $P$, the cooling channel $\Phi(\rho) = P\rho P + \mathrm{tr}((I-P)\rho)P/3$ has dual $\Phi^*(B) = PBP + \mathrm{tr}(PB)(I-P)/3$, which is not the Hamiltonian map $h \mapsto PhP$. Its classical recorded branches lose the relative phase of the two triplets.

A complete twelve-state memory driven by all native C/E labels gives $p = 1/12$, $q = 1/6$, and the dimensionless positive projector average $Y = \sum_k \pi_k P_k$. Its coefficient ratios are $1 : 195.11703757 : 3579.82624571$. A two-angle complete record inverse recovers the matrix with exact cancellation of finite-pulse bias, keeping the mixing and numerical allowances separate. This is a declared feedback service, not a derived physical mass observable. In a further declared interaction, let $F_a$ be the matter lift of the trace-orthonormal six-state current frame. Its Gram operator $C = \sum_a F_a^\dagger F_a$ supplies the complete instrument

$$M_0 = \cos(t\sqrt{C}), \qquad M_a = iF_a \frac{\sin(t\sqrt{C})}{\sqrt{C}}, \quad a = 1, \ldots, 12.$$

Its coarse jump/no-jump effects are gauge covariant; individual adjoint indices are not gauge-invariant species labels. Both components of each Higgs tensor determine its invariant left/right supports. Gating the same memory events with these two jump records gives sector-dependent coefficients without empirical mass inputs. None of the complete geometric catalog $t \in \{\pi/5, \pi/3, \pi/2\}$ reproduces the charged hierarchies; the zero coefficient at $t = \pi/2$ is retained. The same tensors attach these coefficients to the finite Weyl action, and fixed-coordinate detectors invert a separately declared static mean-feedback service at finite angle. Neither calculation supplies a compiled combined history, physical poles, or an interacting continuum limit. The complete replay entry is `model/common_source/native_family_protocol.py`. Every normalized projector average has $Y_s \succeq 0$, $\mathrm{tr}\, Y_s = 1$, and $1/3 \le \lambda_{\max}(Y_s) \le 1$. With the common canonical kinetic normalization, unit-modulus neutral Higgs matching tensors, and one common Higgs norm $v$, its three family action singular values, counted once per family, sum to $v$ in every sector; the ratio between sector maxima is at most three. This restricts that normalized feedback prescription, not the finite gauge theorem or other repair laws. Unconditional event-duty pulses can change the trace and must be derived from their complete correlated record law, without normalizing on the pulsed branches.

The self-contained calculations in `model/observer_particles/` reconstruct the response, representation projectors and interaction tensors from repository-local source operators. Their complete receipts distinguish exact rational or algebraic identities from numerical residuals. They use no measured particle masses. They supply neither a physical chiral field net nor a source-selected Higgs vacuum, Yukawa matrix or mass pole. Calibrated mass fits and rejected mass formulas retain their separate scientific status. A relaxation gap, finite cavity frequency or commuting internal modular offset requires its own mass-shell identification before it can be called a physical rest mass.

### 4.4 Kinetic normalization and physical conventions

For a four-component Dirac convention with positive kinetic metrics,

$$Y_{s,\mathrm{can}} = Z_L^{-1/2} Y_s Z_R^{-1/2} / \sqrt{Z_H}.$$

A literal all-left Weyl bilinear instead has a transpose on the left normalization factor. Conversion between these conventions must precede comparison of coefficients. The up and down sectors share $Z_Q$ because their left fields belong to the same weak doublet.

A record matrix can change a coordinate without changing the mass. For example, $Z_L = Z_R = R > 0$, $Y_s = g_s R$ gives

$$Y_{s,\mathrm{can}} = g_s I/\sqrt{Z_H}, \qquad \det(pR - hg_s R) = \det(R)(p - hg_s)^3.$$

Thus a hierarchical Gram matrix by itself need not generate a hierarchical physical vertex. Eliminating an invertible heavy block in a source kernel also changes its Higgs derivative:

$$\partial_H(A - BD^{-1}C) = A_H - B_H D^{-1}C - BD^{-1}C_H + BD^{-1}D_H D^{-1}C.$$

Both the kinetic and Higgs-dependent parts must come from the same source.

Species are identified by the represented charges, Casimirs and interaction channels, with conjugate-charge and chirality conventions stated. Within a species, sorting a nondegenerate singular spectrum is a permissible naming convention for mass eigenstates; it is not a derivation of three families or their weak-current couplings. For canonical Dirac matrices, write $U_{sL}^\dagger Y_s U_{sR} = \mathrm{diag}(y_s)$. The charged-current matrix is $V = U_{uL}^\dagger U_{dL}$, up to the corresponding ordering and rephasing conventions. Generation need not be a conserved quantum number. A physical CP test uses a rephasing-invariant quantity such as the Jarlskog invariant, not the presence of a complex matrix entry [70, 21].

In the executable all-left convention, a directed complex family Yukawa appears as $\overline{Y}$, with the reverse block its transpose. The creation coefficient is $\epsilon \otimes \overline{K}$. Real symmetric tests cannot detect a mistaken conjugation, so the finite checks include complex matrix units and all branches of the 180-dimensional Nambu representation. Its supplied spatial Weyl/Higgs background is distinct from a source-derived interacting field.

### 4.5 Reusable finite kinematic and interaction identities

The strengthened finite family interface selects rank three under its single-object, faithful-action and operational-cost hypotheses. A separate conditional argument from intrinsic CP capability and weak-sector UV completeness gives $3 \le N_g \le 5$. Neither statement attaches a physical family field or excludes extra light sectors. The ten-row exterior census follows from the 32 subsets of five modes: bidegrees $(c, w)$ have multiplicity $\binom{3}{c}\binom{2}{w}$ and charge $-2c + 3w$. Removing vacuum and top degree gives ten bidegrees. A supplied anomaly-free chiral mask and central weights recover the same six-element kernel. Partition readout can transport this dictionary under declared maps; it does not select the maps, physical parity or laboratory charges.

With a declared scalar quartic, breaking scale and gauge couplings, the finite electroweak-breaking theorem identifies the vacuum shell, its one-dimensional stabilizer, three gauge-orbit Hessian zero directions and the tree gauge-mass ratio. The slot dictionaries then give photon charge $eQ$, neutral shape $g_Z(T_3 - Q\sin^2\theta_W)$, and charged shifts $g_W/\sqrt{2}$ on doublets, zero on singlets. These are identities of the declared infinitesimal action and pairing, with supplied constants. They neither construct physical current operators nor supply mixing, decay amplitudes or a Fermi limit.

On a declared Lorentz module and mass shell, free plane waves persist under the declared translation flow, and their phase along their own frame has proper-time derivative $m$. For $m > 0$ the internal-clock periods are $(2\pi/m)\mathbb{Z}$; the zero-mass clock is constant and has no fundamental period. This is conditional kinematics, without a source-selected mass or clock. Likewise, for a declared complex algebra with $[X_i, X_j] = iq\epsilon_{ijk}B_k$, the Pauli identity is

$$(\sigma \cdot X)^2 = (X \cdot X)I - q\sigma \cdot B.$$

In the stated orbital/spin normalization it reads as $g = 2$. The identity does not construct a gauge potential, electron, anomalous moment or physical coupling.

# 5 Uncalibrated response laws and their exclusions

## 5.1 Complete memory records and normalized feedback

The complete twelve-state memory uses the native events with probabilities $p = 1/12$, $q = 1/6$, a specified reset state and declared rotations. All 128 native raw branches are combined with all twelve old memory states; conditioning on successful events is not substituted for this 1,536-branch law. The stationary projector average

$$Y = \sum_k \pi_k P_k \succeq 0, \qquad \operatorname{tr} Y = 1$$

has coefficient ratios $1 : 195.11703757 : 3579.82624571$. A two-angle record inverse cancels the finite-pulse bias exactly. The resulting numerical allowance is conditional on its floating-point and mixing estimates; it is not an outward-rounded, finite-shot or physical uncertainty certificate.

Gating this memory with the complete thirteen-outcome matter instrument and both components of each Higgs tensor gives different sector coefficients. The entire declared geometric catalog $t \in \{\pi/5, \pi/3, \pi/2\}$ fails the charged hierarchy comparison. Its zero coefficients are retained. A separate finite-angle inverse reconstructs a declared static mean-feedback Hamiltonian from fixed-coordinate preparation and detector probabilities. Averaging the memory and executing a correlated quantum/classical history are different operations; neither calculation proves their dynamical equivalence.

**Proposition 5.1** (Normalized-feedback scale bound)**.** *For a probability average of rank-one projectors on a three-dimensional family space, $Y \succeq 0$, $\operatorname{tr} Y = 1$, and $1/3 \le \lambda_{\max}(Y) \le 1$. With common canonical kinetics, unit-modulus neutral Higgs tensors and a common Higgs norm $v$, the three family action masses in each sector sum to $v$. Any two sector maxima have ratio at most three.*

*Proof.* The nonnegative eigenvalues sum to one, so their largest lies between their mean and their sum. The specified Higgs tensors multiply each family singular value by $v$. Apply the bounds to both sectors. □

This is a sector-normalization obstruction under the common-coupling and canonical-kinetic hypotheses. It does not bound a sector's smallest eigenvalue away from zero and therefore does not exclude arbitrary within-sector hierarchies. A sector-dependent overall strength or a different physical kinetic response changes those hypotheses and needs its own derivation. CP and mixing are separate questions: Yukawa matrices simultaneously real in one weak basis have zero CP-odd invariant, while a common matrix up to sector scalars has aligned left singular frames and trivial charged-current mixing for a simple spectrum. Different real matrices can mix, and general complex positive trace-one matrices need not have vanishing CP violation.

Unconditional event-duty pulses change this normalization. They must use the complete correlated record law. In particular, for reset projector $P_2$, the C-duty response is

$$Z_C = Y - (1 - p)P_2,$$

rather than $pY$. The always/C/E/CE policies have traces $1, p, q, pq$. Their sector-maximum spans in the declared implementation are approximately $1.0233, 2.0572, 1.1528, 2.3695$; the E and CE

responses have rank two. These calculations exclude those policies as the charged mass prescription. They do not exclude every repair law on the carrier.

### 5.2 Quadratic response and the affine-family obstruction

For the native quadratic comparison, put

$$a = \log(25/8), \quad b = \log(8/5), \quad J = a - b, \quad \Omega = a + b.$$

The three eigenvalues of its family matrix $F$ are

$$\frac{61}{450}, \qquad \frac{41}{225} \pm \frac{7\sqrt{5}}{750}.$$

The direct positive prescription has ratios approximately $1 : 1.1903 : 1.4982$, far from a charged hierarchy. Allowing arbitrary complex affine coefficients does not repair this class.

**Proposition 5.2** (Affine quadratic-family exclusion)**.** *Let $Y = uI + wF$, $u, w \in \mathbb{C}$, and let $m_{\min} \le m_{\mathrm{mid}} \le m_{\max}$ be its singular values. With $\varphi = (1 + \sqrt{5})/2$,*

$$m_{\max} \le \varphi^2 m_{\mathrm{mid}} + \varphi m_{\min}.$$

*Consequently this class cannot reproduce the charged-lepton comparison windows used here.*

*Proof.* For ordered eigenvalues $f_1 < f_2 < f_3$ of $F$, the complex points $z_i = u + wf_i$ are collinear. Their adjacent spacings have ratio $\varphi^{-1}$. Solving the affine relation for each endpoint and applying the triangle inequality bounds either endpoint modulus by $\varphi^2$ times either of the other moduli plus $\varphi$ times the remaining modulus. The middle point is a convex combination of the endpoints. Applying these bounds to the largest modulus proves the stated sorted bound. Conditional on $m_{\mathrm{mid}}/m_{\min} \le 207$, it gives $m_{\max}/m_{\min} \le 543.551$, whereas the broad pole-ratio comparison requires at least 3470. This numerical bound is not uniform over all affine coefficients without the middle-to-light ratio constraint. □

This result concerns a fixed quadratic action and its affine family. Its anticommutator pole locations are state-independent; an occupation-energy versus fermionic-gap factor of two, a common clock rescaling or a basis change cannot change their ratios. Interactions that genuinely change the kernel require a separate calculation.

The pole-ratio windows $[206, 207]$ and $[3470, 3485]$ are only a diagnostic target for these dimensionless tree coefficients until a physical matching map is supplied. For example, the adopted $\overline{\mathrm{MS}}$ comparison at $M_Z$ has lepton ratios about 210.7 and 3578, rather than the pole values about 206.77 and 3477.4 [19]. The affine inequality still excludes that comparison, but its numerical bound must use that scheme's middle-to-light ratio. Ratios do not remove all scale and scheme dependence.

Other explicit laws also fail: a five-memory law gives ratios near $1 : 30.84 : 452.62$; a two-rotation law gives $1 : 382.23 : 5909.43$; the admitted auxiliary-dressing class bounds its two smallest subthreshold lepton poles by a ratio below 32. These are specified model exclusions. A near numerical ratio obtained by searching many prescriptions is not a blind prediction; the selection ancestry and all admitted alternatives must be retained.

### 5.3 Full repair and physical propagation

An exact decision experiment first tests a local defect candidate in the unchanged total-load-two native C/E domain. All 720 raw outcome maps are retained on its 78 configurations. Twelve transient double-occupation classes feed one closed irreducible binary class of dimension 66. A scalar configuration invariant preserved by every admitted native branch is therefore constant. Under the separately specified independent uniform-seam/fair-coin schedule, a double load survives $n$ attempts with probability $(5/6)^n$, has mean lifetime six attempt counts, and splits rather than moving as a protected defect. In the uniform binary stationary law the nonzero-current probability is 5/33, while every event has zero drop of $\Phi = \frac{1}{2}\sum_p n_p^2$. Edge-disagreement costs can both rise and fall. Thus motion and repair activity are not that potential decrease. Removing the five seams at one port creates closed classes of sizes 11 and 55 only by changing the allowed schedule. These are configuration-sector results, not an exclusion of quantum, history or enlarged-carrier defects.

For this same collision candidate the full quantum instrument gives the stronger exact identity

$$\sum_h K_h^\dagger P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$$

for either C or E, including zero-probability outcomes in the retained menu. There is no binary-to-collision feed. With the stated independent uniform attempt schedule, arbitrary inputs entangled with an untouched reference therefore have collision weight $(5/6)^n$ times their initial weight. No persistent quantum subspace is supported wholly on these collisions under that unread evolution. This does not classify other quantum or history sectors.

The actively recovered matching code has a different contract. Without recovery, the uniform-seam C/E maps take its normalized code projector outside the code with probabilities 31/108 and 55/108, respectively. With the declared recorded recovery, its logical state and references are preserved while the complete record law is independent of the logical input. A fixed cost of those records alone cannot split logical-state masses. This does not exclude a common energy offset. The fixed matching table, source-derived equivariant code and a new nonscalar correlation observable cannot be silently identified. The native invariant, localized excitation and common energy/clock attachment remain explicit obligations of any defect route.

Passive stationary fixed-reference schedules have an exact rank-two Schur response obstruction, including eight zero crossing weights among the 22 seams. Positive whitening preserves the null direction. The uncorrected recorded 66-state channel has no invariant fifteen-state work code in the tested class. Its ten-mode response conditioned on an incident event has operator norm $\lambda_C = 7/10$ or $\lambda_E = 2/5$. For a unitary $U$ and any linear map $J$,

$$\|UJ - JR\|_{\mathrm{HS}} \geq (1 - \lambda)\|J\|_{\mathrm{HS}}.$$

Thus it has no nonzero exact direct intertwiner to the unmeasured closed quadratic Weyl evolution. The theorem does not exclude a field coupled to detectors, an environment or interactions; a complete dephasing-channel control realizes the same uniform contraction. Such a bridge must match the complete record and continuation law. Actively corrected, copied, controlled and refinement-dependent constructions have separate domains.

Collective charge operators can be reconstructed from shared records without being a physical fermion field. The free spatial implementation has conditional Weyl–Higgs dispersion on a supplied background; its cubic staggered continuum contains two Dirac tastes and both chiralities. A restricted momentum comparison does not prove a unique physical continuum. Likewise, a free wedge modular/stress calculation does not prove physical entropy stationarity or the interacting

Einstein construction [9]. The source program must connect these separate calculations on one admissible field and observer history.

Response diagnostics now separate supplied-action spectroscopy from actual record reconstruction. The finite interacting CAR response determines its inverse and frequency derivative from all Lehmann residues, with canonical sum-rule and positivity admission. Kinetic and mass forms transform together under a change of interpolators; raw matrix eigenvalues are not invariant mass data. A separate charge-resolved spectral census of the supplied 180-mode Weyl/Nambu symbol retains signed and dark modes, unresolved frequency lists and numerical rather than certified exact residue ranks.

For one existing charged $L/e^c$ pair, a finite neutral-reference experiment reconstructs the complex two-mode transfer from complete detector/clock records. Charge-compatible source pulses prepare a fixed-total-charge-zero reference, and all measured continuation branches remain. On a supplied principal logarithm window and canonical quadratic response class, the inverse recovers the generator, residues and joint z-axis momentum and neutral-Higgs multiplier derivatives. The retained 304 contexts have 4,864 terminal outcomes. Withheld settings and a second duration give generator residuals at most about $6.40 \times 10^{-8}$ per unit of reconstructed clock coordinate and complete-output distances about $4.36 \times 10^{-8}$. Both durations repeat the same elementary source word; their composition defect is about $2.23 \times 10^{-16}$. These are numerical diagnostics, not physical error certificates. Phase-sensitive controls detect opposite coupling signs hidden by Rabi marginals, and withheld momenta detect an otherwise indistinguishable two-hop kinetic control. The alias-window prior is not removed by the two durations.

This reconstructs a component of the existing supplied action. It selects no mass coefficient, introduces no new particle candidate, supplies no dynamical Higgs/current vertex and does not establish an interacting continuum. Its completed coherent-control services and clock remain premises; native attempt, abort and export histories are not replaced by unit success probability. The executable entry points and complete readout contracts are recorded in `docs/research/OPH_FIELD_RECONSTRUCTION.md`.

There is already a finite abelian quantum link with an electric–matter interaction and an exact operator-Gauss sector; it is a restricted four-state code, not the full nonabelian particle theory. The Euclidean overlap calculation on the retained fifteen-state representation goes beyond a free symbol: it checks the modified chiral projector and gauge covariance on a nonconstant nonabelian background, with a stated small-plaquette locality domain. Its native kernel queries retain success and failure branches. These results leave the global chiral measure, reflection positivity or another justified Lorentzian construction, dynamical gauge/Higgs fields and common physical clock open. The naive lattice's eight balanced Weyl nodes are a negative control; deleting them does not prove decoupling. The precise retained domains are documented in `model/common_source/DERIVATION.md` and `model/common_source/chiral_candidate/REGULATOR_ROUTES.md`.

# 6 Flavor constraints and comparison constructions

## 6.1 What a family band determines

A source-executed protected rank-three band is a candidate family carrier. Its identification with three physical generations is additional. If both family factors have the same unbroken irreducible action and the Higgs is family-neutral, Schur's lemma forces $Y_s = y_s I_3$, hence degeneracy. Splitting requires a physical symmetry-breaking response. A declared transport kernel, its centered generator, overlap lines, gaps and cycle phases are useful mathematical data only after their common source and physical channel are specified.

For a positive Hermitian circulant square-root response $C = aI + bR + \bar{b}R^2$, $R^3 = I$, let $b = are^{i\delta}$. If its eigenvalues are the nonnegative square roots of the masses, then

$$Q := \frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{1}{3} + \frac{2}{3}r^2.$$

Indeed, $\operatorname{tr} C = 3a$ and $\operatorname{tr} C^2 = 3a^2 + 6|b|^2$. Thus $Q = 2/3$ exactly when $r = 1/\sqrt{2}$. This is the Koide/circulant identity [14, 17]; it does not select a positive branch, the phase, an absolute scale or a physical response. A supplied GNS balance condition is a premise rather than a consequence of this trace identity. Its retained equal-weight block realization uses an asymmetric choice of blocks abstracted from the measured charged triple. Treating those blocks alike gives a different amplitude and $Q = 1$. Thus the construction realizes the empirical balance condition; it does not independently select Koide's value or the fitted phase $2/9$.

Orientation reversal can send a candidate cycle phase $\delta \mapsto -\delta$. Identifying that operation with physical antiunitary time reversal requires its own intertwiner. A classical cycle affinity is a log probability ratio, not a coherent U(1) phase. Detailed balance of one declared kernel does not rule out phases in another coherent sector or in a controlled extension. A phase inference must use the actual source process and an observable phase readout.

### 6.2 Quark moduli and common-scale comparison

Conditional on the registered source signature containing no typed datum that separates the stated centered-spread orbit, a generic quark package does not define a unique quark mass spectrum. Granting an ordered profile shape, write its centered logarithms as $\sigma_u \widehat{\ell}_u$ and $\sigma_d \widehat{\ell}_d$. Independent positive changes of $(\sigma_u, \sigma_d)$ preserve the certified gauge representations, anomaly traces and invariant Higgs lines. Those certificates therefore leave at least an $(\mathbb{R}_{>0})^2$ fiber in this granted subfamily. Without the shape hypothesis, sector means and shapes are also undetermined.

Representative independence does not select the datum: any constant map on a selected bridge fiber descends to its quotient, whatever its constant value. A target-derived spread retains that ancestry after descent. For the minimal light branch

$$y_u = c_u\varepsilon^6, \quad y_d = c_d\varepsilon^6, \quad \varepsilon = 1/6, \quad \Delta_{ud} = \tfrac{1}{6}\log(c_d/c_u) = t_1/5,$$

the conditional bridge fixes a ratio, not either independent spread. The candidate reciprocal-ray law fails on common-scale Standard Model $\overline{\text{MS}}$ Yukawas at $M_Z$ [19]: its required product of adjacent log-gap ratios is one, whereas the adopted comparison gives approximately 0.83532. That is a rejection on the stated comparison surface, not a global classification of quark sources.

The one-doublet tensors can type a relation between down-quark and charged-lepton channels; they do not require equal coefficients. Under the additional alphabet, measure-balance and register-faithfulness hypotheses, finite enumeration selects $\{1/3, 1, 3\}$. Ranking its six generation assignments against measurements is target-informed. The selected assignment fails the adopted joint FLAG comparison [20]. The assignment $y_b = y_\tau$, $y_s = y_\mu/3$, $y_d = 3y_e$ is the Georgi–Jarlskog texture [62]; its historical ancestry and target-informed ranking are distinct from the conditional alphabet enumeration. The retained six-quark mixed-convention table also comes from a target-conditioned template search and is not an independent common-scale prediction. Similarly, directly identifying two real $A_5$ residual axes with the Cabibbo angle is excluded in that class: the least nonzero angle is about $20.905°$, with sine 0.356822, rather than the measured approximately 0.225. This does not exclude general complex flavor transport or additional dynamical misalignment.

### 6.3 Conditional fixed-pixel and electroweak matching charts

Explicit fixed-$P$ numerical constructions evaluate running-chart coordinates after their flavor law, absolute clock and matching prescriptions are frozen. They are conditional constructions: their discrete choices and some attachments have target ancestry. Freezing them does not erase that ancestry. GeV-valued textures assembled from mixed-scale quark rows are not dimensionless Yukawa matrices at a common scale. A physical comparison must transport all six coordinates, including the top extraction convention, to one scheme and scale, emit the running Higgs expectation value in that scheme, and then use $y_q(\mu) = \sqrt{2}m_q(\mu)/v_{\rm SM}(\mu)$.

The same boundary applies to an electroweak matching stage that keeps only the top Yukawa in a Standard Model truncation. A root of that declared map is not a unique source-selected vacuum. Conditional solutions of $\lambda = \beta_\lambda = 0$ select a critical boundary after its couplings, truncation, clock and matching are supplied. Positive-capacity top bands and fixed-pixel near-agreements do not remove those premises.

**Running content and branch provenance.** The historical transmutation recipe used the MSSM one-loop coefficients $(33/5, 1, -3)$, with $g_1 = \sqrt{5/3}\, g_Y$. The supplied Standard Model content of three fifteen-state generations and one scalar Higgs doublet instead gives $(41/10, -19/6, -7)$ in that convention, or $(41/6, -19/6, -7)$ for $(g_Y, g_2, g_3)$, with $\mu\, dg_i/d\mu = b_i g_i^3/(16\pi^2)$. No superpartner threshold content is derived here. The MSSM packet is therefore an additional benchmark recipe, not the running of the stated matter construction. Correcting it requires a common normalization, threshold/matching specification and recalculation of the closure and transmutation maps; it need not preserve a unification root or the previous numerical agreement.

The near-measurement archived W/Z tuple used $P = 1.63094$. It must not inherit the label of the distinct strict closure branches near 1.6309682 and 1.6309721. Values from those branches, and fitted W/Z reparametrizations, have separate provenance. The legacy absolute normalization $E_\star \simeq 1.22089 \times 10^{19}\,\text{GeV}$ is the Planck energy $\sqrt{\hbar c^5/G}$ using measured $G$; rewriting it as a clock coordinate does not derive that input. The retained source records and their current classifications are indexed in `code/particles/MASS_CANDIDATE_STATUS.md`.

**Criticality is a conditional boundary.** The Higgs/top calculation imports Standard Model running and empirical gauge inputs [23]. The scale $E_\star e^{-\pi} P^{-1/6}$ was proposed after exposure to the implied boundary scale; it is a post-exposure candidate, not an independently selected source boundary. The retained comparison with its supplied 172.1 GeV top reference gives a two-loop curve coordinate 124.7256 GeV for the Higgs and an implied boundary scale $2.9545 \times 10^{17}$ GeV. The older 125.72 GeV curve display does not describe that comparison. The one-loop top range is $[163.0899, 170.7539]$ GeV, so it does not bracket the adopted top reference. These are arithmetic comparisons of declared coordinates, not physical confidence intervals: the Higgs tree coordinate, QCD-converted top and experimental mass conventions still require common matching and theory uncertainties. See `code/particles/runs/calibration/d11_criticality_comparison.json` and `code/particles/runs/calibration/d11_boundary_scale_selection_audit.json`.

**Strong-CP boundary.** Strong CP is independent of the spread and normalization obstructions. The source does not emit the anomaly-invariant $\bar\theta$ or prove that it vanishes. That requires a common-scale physical quark determinant phase and a source law for the QCD topological angle; GeV-valued mass textures supply neither.

### 6.4 Charged shape, determinant and finite-register candidates

On the declared ordered carrier $(-1, x_2, 1)$, with $x_2 = -0.5175863354681689$, two supplied source-support scalars $(\eta, \sigma)$ determine the centered charged log triple

$$\ell_e = -\frac{(3+x_2)\sigma - \eta}{6}, \quad \ell_\mu = \frac{x_2\sigma - \eta}{3}, \quad \ell_\tau = \frac{(3-x_2)\sigma + \eta}{6}.$$

Its sum is zero. The downstream algebraic readout is $m_i = g_e e^{\ell_i}$; neither the displayed carrier coordinate nor the physical source/charged attachment follows from this formula.

**Theorem 6.1** (Charged absolute-scale underdetermination)**.** *Any readout factoring only through this centered log triple and its difference invariants leaves $g_e > 0$ free.*

*Proof.* For every $c \in \mathbb{R}$, $Y_e(c) = e^c \operatorname{diag}(e^{\ell_i})$ has the same centered logs and differences. Its determinant is $e^{3c}$, and its masses scale by $e^c$. Thus the stated data determine a common-shift quotient, without a section selecting the determinant scale. □

An uncentered physical response could supply the affine-covariant anchor

$$A_{\mathrm{ch}}(\ell + c\mathbf{1}) = A_{\mathrm{ch}}(\ell) + c, \qquad A_{\mathrm{ch}} = \tfrac{1}{3}\log|\det M_e|.$$

The common-shift theorem does not exclude such an additional datum. A formal source determinant character must be attached to the physical charged determinant line before this identity becomes a mass law.

**Proposition 6.2** (Twenty-four-register rate no-go)**.** *An oriented register with twenty-four slots, a twenty-four-stage automaton, or a decomposition into twenty-four formal update steps determines no physical frequency, Hamiltonian gap, or mass scale. Indeed, rescaling every physical generator by $H \mapsto \lambda H$, $\lambda > 0$, preserves the register, transition graph, grading, and settled quotient while multiplying all generator gaps and rates by $\lambda$. A repair iteration counter is likewise not an operational clock. A physical rate requires a source-derived clock instrument, calibrated comparison line, and residual bound.*

**Theorem 6.3** (Conditional charged determinant-clock attachment)**.** *Let $L_{24}$ be a one-dimensional source-derived operational clock line with a distinguished calibrated positive gap vector $\ell_{24}$ of norm $\Delta_{24}$. Suppose there is a refinement-natural, quotient-visible norm-preserving line isomorphism*

$$\Theta_e : L_{24}^{\otimes 3} \longrightarrow \det M_e$$

*to the physical charged-lepton determinant line, together with the load-bearing element compatibility*

$$\Theta_e(\ell_{24}^{\otimes 3}) = \det(M_e).$$

*Then the common charged mass scale is fixed by*

$$g_e := |\det M_e|^{1/3} = \Delta_{24}.$$

*More generally, if the element-compatible map rescales norms by a separately derived factor $\kappa > 0$, then $g_e = \kappa^{1/3}\Delta_{24}$. Thus the implication from a calibrated clock-line attachment to the determinant scale is exact.*

*Proof.* The distinguished tensor-cube vector has norm $\|\ell_{24}^{\otimes 3}\| = \Delta_{24}^3$. Element compatibility and norm preservation give $|\det M_e| = \Delta_{24}^3$; take the positive cube root. □

These hypotheses require a source-derived clock line and an element-compatible physical attachment. Counting slots or writing a norm-preserving line isomorphism without the distinguished-vector equality does not construct either.

**Proposition 6.4** (Finite charged-register realization)**.** *The stipulated charged register-and-path class is nonempty. There is an explicit finite algebraic model with register dimensions*

$$(50, 31, 10, 512, 77, 21, 27, 5),$$

*connected transition graphs, normalized tracial states, rank-one event projectors, and the eight declared path weights*

$$\left(\frac{1}{50}, -\frac{1}{31}, -\frac{1}{310}, \frac{1}{512}, \frac{1}{77}, \frac{1}{21}, \frac{1}{27}, \frac{1}{135}\right).$$

*Each noncentral event in $B(H_r)$ admits an accepted/rejected central record dilation in $B(H_r) \otimes D_2$. The graph family is covariant under the sixty proper icosahedral rotations, and normalized path weights are unchanged under inert ancillary stabilization $X \mapsto X \otimes I_k$.*

*Proof.* For each declared connected graph, the diagonal matrix units and both directed units on every edge generate $E_{ij}$ along graph paths and hence generate the full matrix algebra. The normalized trace of a rank-one event is the reciprocal register dimension, and tensor products multiply the two depth-two weights. For an event $P$, the pinching map

$$\mathcal{E}_P(X) = PXP + (I - P)X(I - P)$$

is a unital trace-preserving conditional expectation. Its instrument writes $P\rho P$ and $(I-P)\rho(I-P)$ into orthogonal accepted/rejected pointers in the central $D_2$ factor. Explicit permutation intertwiners preserve graph incidence and ranks for all sixty proper rotations. Finally, $\tau_{dk}(P \otimes I_k) = \tau_d(P)$, and the partial-trace coarse map retracts the inert embedding. These facts construct one model of the declared schema. □

Proposition 6.4 proves formal nonemptiness. It supplies no derivation of the charged source. The eight dimensions, path automaton, coupling degrees, grading, signs, clock, and scalar response are model inputs. The verifier exhausts that automaton, not the physically admissible OPH path space. The construction resolves the abstract event-versus-central-record issue at fixed cutoff. Physical response selection, cofinal screen refinement, and charged-sector attachment are not supplied.

The dimensions, paths, signs, state, scalar response and clock are declared. This proposition proves that their finite schema is realizable; it does not select that schema from the physical source.

**Proposition 6.5** (Conditional charged nature and singularity transport)**.** *Let $M_F$ be the positive face endpoint. Suppose a physical chiral three-family carrier supplies a natural unitary $J_L$, canonical kinetic metrics, and*

$$\frac{v^2}{2}\widehat{Y}_e\widehat{Y}_e^\dagger = J_L M_F^2 J_L^\dagger.$$

*Then its positive left charged response is $J_L M_F J_L^\dagger$. Suppose in addition that the exact renormalized charged kernel is supplied, its singular lines satisfy the declared infrared and regularity conditions, and a finite-register/Dyson map has singularity readout $J_L M_F J_L^\dagger$. Regular invertible field changes preserve the zero set, and a regular Nielsen factorization $\partial_\xi K = A_\xi K + K B_\xi$ makes each simple zero gauge independent.*

*Proof.* The first statement follows from uniqueness of the positive square root. Multiplication of $K$ on the left and right by analytic invertible matrices multiplies its determinant by nonvanishing factors. Jacobi's identity gives

$$\partial_\xi \det K = (\operatorname{tr} A_\xi + \operatorname{tr} B_\xi) \det K,$$

so a simple zero cannot move while the Nielsen factors are regular. These are the standard mixed-fermion and gauge-independence implications used in the pole literature [28, 29]; the stable charged-line infrared qualification follows the infraparticle boundary [30, 31]. □

The displayed square identity and singularity attachment are physical premises. The free witness $K_0(s) = sI - M_F^2$ does not supply an interacting charged-lepton kernel. A determinant-preserving change of three ordered masses by $(2, 1/2, 1)$, with coherently changed mass, Yukawa and pole matrices, preserves the tested algebraic relations unless the endpoint identity $M_F = g_{\mathrm{end}} S_F$ is separately enforced. Existence and internal consistency are weaker than source selection and parameter identification.

**Target-informed response architecture.** The eight-register candidate stipulates its path weights, phase, amplitude and determinant exponent. Its declared response map is a contraction with constant $1.5 \times 10^{-3}$, giving ratios approximately $m_\mu/m_e = 206.7683$, $m_\tau/m_e = 3477.3655$. A segregated comparison is about 84 ppm below the adopted charged triple. Those architecture choices were made with knowledge of the target; a runtime-reference-free evaluation is therefore a target-informed model coordinate. Neither this proximity nor the Koide identity derives the physical source law. A normalized rank-one path weight is not by itself a $\mathbb{Z}_6$ character or a determinant.

The retained inverse comparison has three target-informed spectral coordinates: phase, modulus and scale. The absolute 84 ppm display uses the branch $P \simeq 1.6309720958588974$ and the externally normalized $E_\star$; it is not the relative accuracy of the two mass ratios. The associated ratio comparison reports residuals about $+0.0865$ ppm for $m_\mu/m_e$ and $+0.0790$ ppm for $m_\tau/m_e$. The former lies outside the adopted componentwise one-standard-uncertainty interval of halfwidth about 0.0221 ppm. This interval check supplies neither a joint statistical test nor a physical significance without covariance, matching and model-error information. The comparison and formula-selection ancestry must remain attached to the displayed proximity.

**Electromagnetic scale inversion.** On a declared on-shell transport branch, a common rescaling $m_i \mapsto e^\kappa m_i$ gives

$$P_{\mathrm{lep}}(\kappa) = \frac{1}{3\pi} \sum_i [2 \log(M_Z/m_i) - 5/3], \qquad \frac{dP_{\mathrm{lep}}}{d\kappa} = -\frac{2}{\pi}.$$

An empirical photon endpoint, target-anchored ratios, hadronic payload and scheme bridge can therefore identify $\kappa$. They add data excluded from the common-shift theorem's antecedent list. The retained interval calculation gives central masses approximately $(0.5089, 105.22, 1769.5)$ MeV with comparison intervals containing the adopted measurements. Its anchor-gap uncertainty is correlated with its hadronic payload; treating them independently enlarges the interval. Neither the rectangle nor the payload-coherent interval is source-only. Their interpretation requires the declared higher-order and scheme uncertainties, rather than a physical mass claim from centered data. The higher-order leptonic subtraction uses the measured charged triple; solving back for its

common scale is therefore a calibration consistency check, not an independent falsification target for those masses.

The exact one-loop spacelike Ward response offers another conditional scale attachment. For fixed centered shape $\ell$ and nonzero spacelike momentum, define

$$I(z) = \int_0^1 x(1-x)\log(1+zx(1-x))\,dx, \qquad \mathcal{W}_Q(\mu;\ell) = \frac{2}{\pi}\sum_i I(q^2 e^{-2(\mu+\ell_i)}).$$

Since $I'(z) > 0$, this response is strictly decreasing in $\mu$, with limits infinity and zero. A supplied source-complete endpoint pair and nonleptonic subtraction therefore select a unique common scale. They are additional inputs to the centered-data reduct. Higher-order monotonicity, source ancestry and the clock/atomic transport factors are separate requirements; inverting measured endpoints is calibration.

### 6.5 Conditional neutrino deformation

Fix an isotropic Majorana reference $M_0$, with real $a, \rho > 0$ and cycle phase $\Omega$. Supplied same-label gap and overlap data $g_e > 0$, $0 \le \omega_e < 1$, $e \in \{12, 23, 31\}$, give

$$q_e = \sqrt{g_e(1-\omega_e)}, \quad \eta_e = \log q_e - \tfrac{1}{3}\sum_f \log q_f, \quad \mu_e = \frac{e^{\eta_e}}{\frac{1}{3}\sum_f e^{\eta_f}}.$$

The gaps and overlaps in this comparison inherit a template transport kernel and candidate line lift. Centering removes a common scale; it does not establish their source ancestry or physical family attachment.

**Theorem 6.6** (Fixed isotropic reference plus same-label scalars suffice for the intrinsic deformation)**.** *Fix the isotropic reference $M_0$, equivalently its parameters $(a, \rho, \Omega)$. Assume the same-label gap and overlap scalars are supplied on all realized arrows. Then the full intrinsic neutrino mass-eigenstate bundle factors through the scalar certificate*

$$(g_e, \omega_e)_{e\in\{12,23,31\}}$$

*or equivalently through the centered eta-class $[\eta_e]$. No additional raw matrix payload is needed to form the intrinsic selector, the depressed-cubic spectrum, the ascending gaps, or the intrinsic spectral subspaces. When the singular spectrum is simple and strictly positive, canonical Takagi congruence fixes an ordered column basis up to column signs. If a simple spectrum contains one zero singular value, its zero column retains an arbitrary $U(1)$ phase while the nonzero columns are fixed up to signs. At a degeneracy, including the isotropic $s_0 = s_1$ point, only the degenerate spectral subspace and its projector are fixed. Ascending singular-value sorting does not select the physical normal or inverted mass-eigenstate labels.*

**Theorem 6.7** (Exact principal-branch selector from the centered eta-class)**.** *Fix the cycle sum*

$$\Omega = \psi_{12} + \psi_{23} + \psi_{31},$$

*and a positive weight family $\mu_e$. Define*

$$\mu_{\min} = \min_e \mu_e, \qquad F_{\max} = \sum_e \arcsin\left(\frac{\mu_{\min}}{\mu_e}\right).$$

*If $|\Omega| < F_{\max}$, then on the principal branch $\psi_e \in (-\pi/2, \pi/2)$, the affine energy*

$$A(\psi) = \sum_e \mu_e(1 - \cos\psi_e)$$

*has a unique minimizer. It is given by*

$$\psi_e^\star = \arcsin(\lambda/\mu_e),$$

*where $\lambda$ is the unique solution of*

$$\sum_e \arcsin(\lambda/\mu_e) = \Omega, \qquad \lambda \in (-\min_e \mu_e, \min_e \mu_e).$$

*Proof.* The Euler–Lagrange equations are $\mu_e \sin\psi_e = \lambda$. On the principal branch the scalar function

$$F(\lambda) = \sum_e \arcsin(\lambda/\mu_e)$$

is strictly increasing because

$$F'(\lambda) = \sum_e \frac{1}{\sqrt{\mu_e^2 - \lambda^2}} > 0.$$

Its range on $(-\mu_{\min}, \mu_{\min})$ is $(-F_{\max}, F_{\max})$, so $F(\lambda) = \Omega$ has a unique solution under the stated condition. Strict convexity follows because the Hessian of $A$ is

$$\nabla^2 A = \operatorname{diag}(\mu_e \cos\psi_e),$$

which is positive definite on the principal branch and is positive definite after restriction to the affine plane $\sum_e \psi_e = \Omega$. □

For the selected phases put

$$M(\eta) = \begin{pmatrix} a & \rho e^{i\psi_{12}} & \rho e^{i\psi_{31}} \\ \rho e^{i\psi_{12}} & a & \rho e^{i\psi_{23}} \\ \rho e^{i\psi_{31}} & \rho e^{i\psi_{23}} & a \end{pmatrix}.$$

**Theorem 6.8** (Exact intrinsic spectral cubic)**.** *Let $H = M^\dagger M$. Then*

$$H = dI + T, \qquad d = a^2 + 2\rho^2,$$

*where $T$ has zero diagonal and off-diagonals*

$$\begin{aligned} x_{12} &= 2a\rho\cos\psi_{12} + \rho^2 e^{i(\psi_{23} - \psi_{31})}, \\ x_{23} &= 2a\rho\cos\psi_{23} + \rho^2 e^{i(\psi_{31} - \psi_{12})}, \\ x_{13} &= 2a\rho\cos\psi_{31} + \rho^2 e^{i(\psi_{23} - \psi_{12})}. \end{aligned}$$

*Define*

$$P = |x_{12}|^2 + |x_{23}|^2 + |x_{13}|^2, \qquad Q = \Re(x_{12}x_{23}\overline{x_{13}}).$$

*Then the eigenvalues $\lambda_k$ of $T$ are exactly the three real roots of*

$$\lambda^3 - P\lambda - 2Q = 0,$$

*and the intrinsic squared masses are*

$$m_k^2 = d + \lambda_k.$$

*If $P > 0$, equivalently,*

$$\lambda_k = 2\sqrt{\frac{P}{3}}\cos\left(\frac{1}{3}\arccos\left(\frac{3\sqrt{3}\,Q}{P^{3/2}}\right) - \frac{2\pi k}{3}\right).$$

*If $P = 0$, all three off-diagonal entries vanish, $T = 0$, and $\lambda_0 = \lambda_1 = \lambda_2 = 0$; the trigonometric expression is not used.*

**Numerical mass readout.** The cubic identity is an exact algebraic statement, not a stability claim for its floating-point evaluation. Near a massless locus, subtracting $d+\lambda_k$ loses small positive squared masses; forming $M^\dagger M$ also squares the condition number. For example, at $\Omega = 0$, $\eta = 0$, $\rho = 1$, and $a = 1 + 10^{-8}$, the two small masses are $a - \rho$, although binary64 evaluation of the cubic can round both to zero. The executable mass readout uses a scaled direct singular-value decomposition of $M$, checks the positive Takagi congruence, and retains the cubic calculation only as a diagnostic. These are normwise floating-point residual checks, not interval enclosures or relative-error certificates for arbitrarily small masses. A degenerate block requiring additional congruence resolution is explicitly refused. This numerical repair supplies no physical mass-eigenstate labels or source closure.

**Corollary 6.9** (Intrinsic eta-chain spectral closure)**.** *Once the centered eta-class is emitted at the flavor boundary, the intrinsic neutrino branch emits*

$$s_0 \le s_1 \le s_2, \qquad s_1^2 - s_0^2,\ s_2^2 - s_0^2,\ s_2^2 - s_1^2,$$

*and the corresponding intrinsic spectral subspaces with no PMNS import and no flavor-label leakage. A columnwise Takagi basis up to signs is emitted only on the strictly positive simple-spectrum locus; a simple zero mode retains its phase freedom. The physical normal-ordering assignment is* $(\nu_1, \nu_2, \nu_3) = (s_0, s_1, s_2)$*, while the inverted-ordering assignment is* $(\nu_3, \nu_1, \nu_2) = (s_0, s_1, s_2)$*. The source does not choose between them and therefore does not select a physical ordering.*

The scalar-certificate statement follows because centering determines all $\mu_e$, the principal-branch theorem determines the phases, and the matrix and its spectral projectors then follow. The cubic is obtained by expanding $\det(\lambda I - T)$. These implications do not determine the supplied reference, its absolute scale or the physical charged basis. On the isotropic branch two singular values coincide, giving zero solar splitting. In the anisotropic comparison the template-derived charged basis gives angles incompatible with the adopted oscillation data [27]; this is a failed conditional diagnostic rather than an OPH prediction of PMNS angles. An ordering and a charged-sector intertwiner must be selected before a physical neutrino comparison.

# 7 Electroweak charts, scale and pole attachment

## 7.1 What the two-output chart can fit

Fix positive running couplings $\alpha_2, \alpha_Y$, a Higgs chart value $v$, and a source coefficient $\eta$. In this subsection $v$ uses the conventional $v_{\rm SM}$ normalization. Define

$$\tau_Y(\tau) = -\frac{\tau + 2\eta}{1 + 4\tau^2}, \qquad n_{\rm fib}(\tau) = 1 + \frac{\alpha_2 \tau + \alpha_Y \tau_Y(\tau)}{\alpha_2 + \alpha_Y},$$

$$M_W = v\sqrt{\pi\alpha_2(1+\tau)}, \qquad M_Z = v\sqrt{\pi(\alpha_2+\alpha_Y)[n_{\rm fib}(\tau) + \delta n]}.$$

Here $(\tau, \delta n)$ are chart coordinates, rather than emitted physical self-energies. Any positive target pair in this chart can be fitted by

$$\tau = \frac{M_W^{\dagger 2}}{\pi v^2 \alpha_2} - 1, \qquad \delta n = \frac{M_Z^{\dagger 2}}{\pi v^2(\alpha_2+\alpha_Y)} - n_{\rm fib}(\tau).$$

Substitution proves the exact hit. It is a measured-pair reparametrization.

**Proposition 7.1** (Local nondegeneracy of the electroweak chart)**.** *On $1+\tau>0$, $n_{\rm fib}+\delta n>0$,*

$$\det\frac{\partial(M_W,M_Z)}{\partial(\tau,\delta n)}=\frac{M_W M_Z}{4(1+\tau)(n_{\rm fib}+\delta n)}>0.$$

*Proof.* The first mass is independent of $\delta n$; the diagonal derivatives are $M_W/[2(1+\tau)]$ and $M_Z/[2(n_{\rm fib}+\delta n)]$. □

Local invertibility does not prove that the source has exactly two response coordinates or selects a particular point.

**Definition 7.2** (Finite quotient-path certificate)**.** *After removing gauge, port-label and scheduler presentation coordinates, assume the CP-even charge-preserving response through two returns has exactly two real scalar channels and no off-block term. Require independently specified carrier paths, a primitive response $\lambda=\eta\alpha_U/4$, charged incidence $(1,2/3,1)$, neutral incidence $(1,4/3,2)$, and a diagonal central projector of normalized trace $1/6$ and amplitude $\rho$. Its degree-two subtraction is $(\rho/6)\eta^2$ in both channels. Require contracting charged and uplifting neutral signs, and a strictly convex hypercharge least-norm problem with Gram factor $1+4\tau^2$ and residual $\tau+2\eta$. The paths must exhaust the admitted class; deeper paths and alternative weights must be absent, quotient-trivial or excluded by a positive, target-independent selector gap.*

**Theorem 7.3** (Conditional quotient-transport value law)**.** *Assume one target-independent source branch emits $(\alpha_U,\alpha_2,\alpha_Y,\eta,v)$, with $\rho=(\alpha_2-\alpha_Y)/(\alpha_2+\alpha_Y)$, $\eta=\rho\alpha_U$. Under Definition 7.2, the unique admitted transport is*

$$\tau=-\lambda[1+2\eta/3+(1-\rho/6)\eta^2],\qquad \delta n=\lambda[1+4\eta/3+(2-\rho/6)\eta^2],\quad \lambda=\eta\alpha_U/4.$$

*It gives the displayed chart masses whenever $\alpha_2'>0,\alpha_Y'>0$, where*

$$\alpha_2'=\alpha_2(1+\tau),\qquad \alpha_Y'=\alpha_Y[1+\tau_Y(\tau)]+(\alpha_2+\alpha_Y)\delta n.$$

*Proof.* The primitive normalization, return incidences, central subtraction and signs fix the two polynomials. Strict convexity fixes the hypercharge minimizer. Substitution gives the mass chart, and exhaustion excludes alternative admitted responses. □

The certificate, including its exhaustive path category, is a hypothesis. Assigning weighted paths to the monomials of a desired polynomial does not prove that the physical source selects those paths or weights.

## 7.2 Electromagnetic readout and Maxwell normalization

**Theorem 7.4** (Conditional Ward-projected transport)**.** *Assume a common physical branch supplies the photon and neutral/charged polarization kernels, a Ward projector onto $Q=T_3+Y$, a positive heat-kernel parameter $t_Q(q^2)$, and a finite strictly positive Thomson limit. If $p_n\propto\exp(-t_Q n^2)$ and $g_Q^2=t_Q/(2\pi)$, then*

$$\alpha_{\rm em}^{-1}(q^2)=\frac{8\pi^2}{t_Q(q^2)},\qquad \Delta\alpha(q^2;m_Z^2)=\frac{t_Q(m_Z^2)}{t_Q(q^2)}-1.$$

*For $a_0=\alpha_{\rm em}^{-1}(m_Z^2)$, the Thomson inverse is $a_0 t_Q(m_Z^2)/t_Q(0)$.*

*Proof.* Use $\alpha = g_Q^2/(4\pi)$ on the same branch and take its stipulated Thomson limit. The Ward projection separates the unbroken electromagnetic line; it does not construct its kernel or the heat-kernel law. □

**Corollary 7.5** (Conditional Maxwell normalization)**.** *If this branch supplies a nonzero Maxwell kinetic action with coupling $g_Q$, varying it gives $d{*}F = g_Q^2 {*}J$, with $dF = 0$.*

*Proof.* The field equation is the action's Euler–Lagrange equation; the Bianchi identity follows from the curvature definition. The heat parameter alone supplies no Maxwell action. □

### 7.3 Hierarchy and common screen load

The conditional transmutation branch uses

$$\frac{v}{E_{\mathrm{cell}}} = \exp[-\pi/(2\alpha_U(P))].$$

Its weak-doublet multiplicity is $N_c + 1 = 4$, with additive load $4P$. Finite exterior multiplicity is insufficient to prove a physical equality of screen and electroweak load carriers.

**Theorem 7.6** (Conditional screen/electroweak composition)**.** *Assume twelve equal-trace port-center atoms, the integer total-twelve counting grammar, normalized readback cost, oriented $(12, 30, 20)$ icosahedral incidence and the stated refinement cocycle. An equal-weight load $X$ has local readout $X/12$. For $X = \log(N/\pi)$, cell entropy $P/4$ and multiplicity four, the defined screen composition is*

$$\Gamma_{\mathrm{scr}} = \frac{P}{12}\log(N/\pi).$$

*If an independently supplied correctable-record capacity $N$ and physical common-load map identify this with $\log(E_{\mathrm{cell}}/v)$, then*

$$\mathcal{B}_{\mathrm{EW}}(P, N) := \alpha_U(P)\log(N/\pi) - 6\pi/P = 0, \qquad N = \pi\exp[6\pi/(P\alpha_U(P))].$$

*Proof.* The declared counting cost $H(q) = 12 + \sum_p (q_p - 1)^2$ has its unique all-one minimum with gap two on the integer total-twelve fiber. The oriented incidence supplies inverse pairing, the proper $A_5$ action and a rank-three frame with $G^2 = 4G$ [6]. Equal-weight additivity gives $X/12$. Multiplying the three stated factors gives $\Gamma_{\mathrm{scr}}$. The additional physical identification and transmutation relation give the last equation. □

The integer Lyapunov cost, normalized readback cost and A3 Hessian are different objects. This local lineage theorem does not construct the federation nerve, identify a primitive observer or derive a global support screen.

**Theorem 7.7** (Defined bridge-map fixed point)**.** *For fixed $P$ and $0 < \lambda \le 1$, the map*

$$\mathcal{C}(P, x) = (1 - \lambda)x + \lambda\frac{6\pi}{P\alpha_U(P)}$$

*is a contraction with unique fixed point $x_\star = 6\pi/[P\alpha_U(P)]$. Therefore $\mathcal{B}_{\mathrm{EW}}(P, \pi e^{x_\star}) = 0$.*

*Proof.* Its derivative is $1 - \lambda$; solve $x = \mathcal{C}(P, x)$. □

This is a defined map, without a proof that it is the physical direct record-capacity map. Finite-presence and Poisson reserve factors, $1 - P/24$ and $e^{-P/24}$, require different composition premises. Neither follows from the contraction or from their shared local expansion.

**Proposition 7.8** (Declared-map RG/Higgs square)**.** *If both routes through source and Higgs quotients are defined to preserve the same tuple* $(P, N, \Theta, F_{\mathrm{crit}} F_{\mathrm{EW}}(P, N))$*, with* $\Theta = (N/\pi)^{-P/12}$*, their comparison defect is zero.*

*Proof.* Both routes return the same tuple by definition. □

This is a declared-map packaging identity, not a construction of source-derived comparison maps. The current packet supplies no OPH result about Higgs naturalness. Finite completions sharing the stated antecedents can instead have defect one. A commuting definition is not a calculation that physical relevant deformations preserve the weak hierarchy.

## 7.4 Operational scale and source Hamiltonian

**Proposition 7.9** (Conditional physical clock attachment)**.** *If the same strict source branch emits the dimensionless energy* $\varepsilon_{\mathrm{clk}} > 0$ *of a declared clock transition whose physical frequency* $\nu_{\mathrm{clk}}$ *defines the operational unit, then, with* $h_{\mathrm{P}} = 2\pi\hbar$ *Planck's constant,*

$$E_\star = \frac{h_{\mathrm{P}} \nu_{\mathrm{clk}}}{\varepsilon_{\mathrm{clk}}}.$$

*Equivalently,* $\gamma_\star = \ell_\star \nu_{\mathrm{clk}}/c$ *gives* $E_\star = \hbar\nu_{\mathrm{clk}}/\gamma_\star$*. An operator-norm perturbation of at most* $\epsilon$ *changes an isolated selected clock gap by at most* $2\epsilon$*. This is a conversion and stability theorem; the required physical reference transition and SI binding are absent.*

**Proposition 7.10** (Exact finite source-Hamiltonian gap)**.** *Let* $D_\sigma$ *be the sign-twisted seam derivative on the connected observer-visible finite local domain, and let* $H_\sigma = D_\sigma^* D_\sigma$ *use the declared unit-counting measure. The signed-incidence formula makes* $H_\sigma$ *integer symmetric and positive semidefinite. A kernel vector would supply a sign-consistent coloring on each connected component. The declared reversing seam transport frustrates every triangle, so the connected complex has no such coloring and the signed incidence rank is* 8,662*. Therefore* $\ker H_\sigma = 0$*. With maximum degree* 12*, the determinant bound gives*

$$\lambda_{\min}(H_\sigma) \geq 24^{-8661} > 0.$$

*A sparse numerical refinement gives* $\lambda_{\min} = 0.11753670336118684$ *with relative residual* $1.49 \times 10^{-14}$*. The decimal is measured rather than certified. The exact statement is strict positivity and the displayed floor.*

The determinant floor follows because the positive integer determinant is at least one and every eigenvalue is at most 24. The numerical decimal has a residual check, not an interval enclosure. This gap does not select a physical field fiber, laboratory transition or determinant attachment.

For an isolated cesium ground manifold, set $P = P_3 + P_4$, $Q = 1 - P$, with $\dim V_3 = 7$, $\dim V_4 = 9$, and define

$$\mathcal{F}_P(z) = P(\widehat{H} - z)P - P\widehat{H}Q[Q(\widehat{H} - z)Q]^{-1}Q\widehat{H}P.$$

**Proposition 7.11** (Cesium channel scalarization and monotone clock roots)**.** $z \in \operatorname{spec} \widehat{H}$ *exactly when* $0 \in \operatorname{spec} \mathcal{F}_P(z)$*, with matching multiplicities. Rotational invariance and the multiplicity-free decomposition* $V_3 \oplus V_4$ *force* $\mathcal{F}_P(z) = f_3(z)P_3 + f_4(z)P_4$*, and* $\mathrm{d}\mathcal{F}_P/\mathrm{d}z \preceq -P$ *gives* $f_F'(z) \leq -1$*. One sign bracket per channel therefore isolates exactly one level, and the clock normal form is* $H_{\mathrm{clock}}^{\mathrm{eff}} = E_0 P + \widehat{a}_{\mathrm{Cs}}\, \mathbf{I} \cdot \mathbf{J}$ *with* $E_0 = (9E_4 + 7E_3)/16$ *and* $\varepsilon_{\mathrm{Cs}} = E_4 - E_3 = 4\widehat{a}_{\mathrm{Cs}}$*.*

The Schur complement preserves the isolated spectrum; Schur's lemma scalarizes the rotational blocks. Differentiating the resolvent gives $\mathcal{F}'_P \preceq -P$, proving the stated root uniqueness after sign brackets are supplied. No atomic Hamiltonian is derived by this reduction. The electromagnetic convention, electron ratio, nuclear packet, scalar evaluators, refinement tails and target-independent provenance are needed to calculate the gap. Source extensions differing in an unfixed parent can give different unique gaps.

**Proposition 7.12** (Operational scale metrology)**.** *In SI units* $E/\mathrm{GeV} = [E/(h\nu_{\mathrm{Cs}})]\,[h\nu_{\mathrm{Cs}}/\mathrm{GeV}]$*, and the second factor is an exact unit conversion. A source-only numerical GeV mass is therefore equivalent to a source-only dimensionless operational clock ratio. Replacing cesium by another clock species adds the burden of predicting that clock's frequency ratio to the defining cesium transition.*

A clock-gap coordinate chosen to reproduce a gravitational coupling is a calibration checksum. Source-action maximum entropy gives a unique law only after its physical operator basis and source moments are supplied; that uniqueness principle does not supply the missing moments.

## 7.5 Renormalization transport and physical singularities

**Proposition 7.13** (Unique frozen RG and threshold transport)**.** *Let the canonically normalized coefficient vector* $X(\mu)$ *obey a locally Lipschitz beta system on each interval of a frozen finite threshold list. Assume that the maximal solution from each incoming matched value exists and remains finite on the full closed interval, and that each matching output lies in the domain of the next beta system. Also assume deterministic matching maps, fixed loop order, field content, renormalization scheme, threshold locations, and decoupling order. Then one source initial condition determines one low-energy coefficient vector. If* $L_j$ *bounds the beta-system Lipschitz constant and* $K_j$ *bounds threshold map* $j$*, perturbations obey the corresponding product of* $e^{L_j \Delta t_j}$ *and* $K_j$*, plus the declared truncation and matching remainders.*

*Proof.* Picard–Lindelöf gives local existence and uniqueness; the stated full-interval non-blowup and domain hypotheses extend the unique solution to every declared threshold. Grönwall bounds perturbations on each interval; applying each deterministic Lipschitz threshold map and inducting through the ordered list gives the stated composition. □

**Theorem 7.14** (Conditional complex-pole theorem)**.** *For* $B = W, Z, H$*, let* $\Gamma_B^{\mathrm{phys}}(s,\xi)$ *be the BRST-complete physical inverse two-point block, including every declared mixing field after the Ward/Slavnov–Taylor projection, analytically continued to a declared Riemann sheet, and set*

$$D_B(s,\xi) = \det \Gamma_B^{\mathrm{phys}}(s,\xi).$$

*Assume a reference determinant has exactly one simple physical zero inside a certified contour, no zero lies on the contour, and the full determinant obeys* $|D_B - D_{B,0}| < |D_{B,0}|$ *there. In the neutral sector also assume the Ward-protected photon line has been separated from the massive eigenvalue. Then the enclosed pole is unique and stable. With the exact convention*

$$s_B = \left(M_B - \frac{i}{2}\Gamma_B\right)^2, \qquad M_B > 0, \quad \Gamma_B \geq 0,$$

*it determines one mass and width. If a Nielsen identity has the form* $\partial_\xi D_B = C_B^{(\xi)} D_B$*, the simple pole is gauge-parameter independent [29, 38]. For* $W$ *and* $Z$*, interpretation as a physical resonance additionally requires a dressed BRST-invariant current amplitude whose Laurent coefficient at that pole is nonzero; for* $H$*, the analogous scalar-amplitude coupling is a separate condition. A simple zero of the inverse-propagator determinant alone does not establish either coupling.*

*Proof.* Rouché's theorem preserves the number of enclosed zeros under the certified perturbation. Simplicity gives the implicit-function stability bound and a locally simple determinant zero. Differentiating $D_B(s_B(\xi), \xi) = 0$ and using the Nielsen identity gives $ds_B/d\xi = 0$. An invertible analytic field redefinition multiplies $D_B$ by a nonzero analytic factor and therefore preserves the pole set. □

The populated physical self-energy kernels, analytic-sheet and contour certificates, widths, residues and scheme uncertainties are absent from the two-output mass chart. A label such as "pole mass" in a schema does not supply these objects.

**Theorem 7.15** (Source-selection and physical-pole composition)**.** *Suppose the admitted source maps are fully specified modulo gauge and trivial reparametrization. Assume the physical readout is constant on that class or a target-independent selector has one minimizer with a positive gap. Require an acyclic dependency graph excluding measured masses and calibrated proxies from every source ancestor, including the specification of the maps. If one source root and physical clock emit the canonical coefficients, the quotient-path and criticality split certificates hold, and the RG and physical-pole hypotheses above are satisfied, then the composition determines a unique enclosed pole triple* $(s_W, s_Z, s_H)$.

*Proof.* The class/selector hypotheses identify one readout. Source separation follows from its complete ancestry. The unique root and scale determine one coefficient packet, the RG theorem transports it uniquely, and the pole theorem supplies one stable physical singularity per sector. □

The displayed constructions do not satisfy this conjunction. Generic symmetry and analyticity permit families such as $\tau = -c\eta^2$, $\delta n = d(1-\rho)\eta^2$, and criticality shifts $(\pi_y, \pi_\lambda) \mapsto (\pi_y + a\eta^N, \pi_\lambda + b\eta^N)$, with different outputs. Even a fixed running mass allows the different kernels $s - m_R^2$ and $s - m_R^2 - \epsilon$. Thus a chart, analytic dependence and channel counting cannot select a physical pole. Numerical residuals must be attributed to separate source, truncation, matching, clock and experimental uncertainties. Scale variation is not a source covariance, and componentwise intervals alone do not specify a joint likelihood.

# 8 Hadrons, records and physical interpretation

## 8.1 Hadronic backend

A source-only hadron prediction requires a source QCD parameter map and quotient ensemble, a vacuum/transfer construction, physical currents and spectral observables, and controlled continuum, volume, light-quark and sampling errors. A two-current spectral measure can feed running-alpha and vacuum-polarization calculations; it cannot supply four-current light-by-light or transition amplitudes by itself. Published hadronic payloads are separate empirical inputs.

For a declared positive Euclidean spectral representation,

$$C_X(t) = \int e^{-Et}\, d\rho_X(E)$$

relates correlators to energy levels. Extracting a stable-channel mass also requires excited-state and finite-volume bounds, an operational scale and a continuum extrapolation. The seeded $2+1$ family and deterministic sampling contract are execution specifications. The retained surrogate uses proxy pion and nucleon masses in constructed correlators. Its receipt-to-evaluation replay tests infrastructure, without supplying production unquenched lattice QCD, Dirac solves, baryon contractions or source-derived hadron masses.

### 8.2 Observer records and particle identity

The finite record theorem constructs a central public record algebra under its compatibility hypotheses. Given a declared algebra/state, instrument and two-wing representation, the measurement theorem gives Born probabilities, the selected Lüders conditioning, repeated-read stability under support-preserving evolution, and the stated CHSH bound [5, 4]. Those results do not identify an arbitrary internal memory state with a particle or prove global observer continuation.

The finite CAR construction describes quantum excitations of a specified source action. No identification as topological defects is established. A physical species identification needs its spin/statistics, gauge charges, propagation, interaction channels and observable spectral response to agree on one field theory. Persistent record tokens can instead be stitched without an implementation-ID rule under the hypotheses in the appendix. That theorem establishes continuation in affine event charts. It does not supply a geodesic action, a mass shell or a physical particle detector.

### 8.3 Antiparticles and asymmetry

A physical chiral-field attachment includes conjugate antiparticle states. A cosmological matter asymmetry additionally requires a physical CP-odd current, its state and generator, a clock, thermal history and charge map. For an antisymmetric CP-odd winding cocycle $\omega$, its candidate rate has form

$$\dot{\Theta}_R = \theta_0 \sum_{q,q'} p(q,T) L_T(q,q') \omega(q,q').$$

A CP-symmetric state and generator make terms cancel against their CP images, giving zero. A settled quotient or an iteration count supplies neither a nonzero physical current nor its rate. The finite anomaly coefficient $k_R = \sum_{\psi\,\mathrm{LH}} r_\psi 2T_2(R_\psi)$ is conditional on the physical charged-field and record-charge assignment. It is not a baryogenesis prediction.

## 9 Requirements for a source-derived spectrum

A physical effective description is a claim about compatible observer statistics and interventions. It need not be a fundamental field or spacetime placed behind those observations. The construction must supply one fixed dictionary for preparations, clock readings, complete detector outputs and subsequent responses, with controlled error over its stated domain. Agreement of eigenvalues or of a few two-point values alone is weaker than that contract.

A complete construction must determine the charged chiral inverse propagator $\Gamma^{(2)}(p;H)$ from the same repair/memory process that produces the observer records. Its Higgs derivative, kinetic forms, background and clock must be derived together. The observer's detector law must identify the physical coefficients or singularities with the empirical mass packet withheld. A continuum refinement then needs the claimed chirality and species content, controlled interactions and an appropriate mass scheme. For confined and charged sectors this includes confinement and infrared qualifications rather than an assumed isolated free-particle pole.

The next calculations have the following dependency order:

1. Specify the carrier/code, state, actual repair and coherent operations, scheduler, controls, access and complete records. Test the seventeen/eighteen current and rank-fifteen obstructions in that exact domain, and identify the spatial and internal actions on it. An additional control, bath or transport compensation remains a premise until implemented and justified.

2. Extract kinetic derivatives, charged gaps and Higgs vertices jointly from one continuing finite process at several momenta and one declared clock convention. Preserve the shared quark kinetic form, complementary modes, unsuccessful detector outcomes and continuation after measurement. This finite test can precede a complete interacting continuum theorem.

3. Test a source mechanism that escapes an explicitly identified normalization or affine-family hypothesis. Record-persistence, localization, radiative and coherent phase constructions are possible classes to examine, not an exhaustive list. A hierarchy calculation and a physical CP/mixing calculation have distinct acceptance tests; a raw nonzero phase is insufficient. Failure excludes the specified candidate or proved class, not all mechanisms on the carrier.

4. Establish compatible locality, state/vacuum, chiral measure and anomaly control, dynamical gauge interactions and controlled volume, spatial and occupation limits for that regime. A finite regulator or free-field success does not discharge this stage. Each additional threshold changes the running and matching obligations.

5. Freeze the source, observation map, calibration packet and uncertainties before the validation comparison. Retain a hash, date, predicted observables, decision rule, all tested alternatives and prior target exposure. Compare ratios within a stated scheme/scale, then sector strengths, mixing and the operational absolute scale. Nine charged Yukawas and the four independent CKM parameters provide a charged-spectrum test; they do not by themselves validate the full interacting observer theory. Independent interaction, neutrino and clock tests retain their own obligations.

Registration after exposure to known masses makes a reproducible postdiction; it does not make the architecture historically target-blind. Numerical replay tolerances, matching/truncation errors and experimental uncertainties must be distinguished. A failed comparison is reported for the frozen rule and its hypotheses; changing the rule creates a new recorded candidate.

Several errors can make a numerical comparison appear stronger than it is. A selected twelve-dimensional subspace need not exhaust the available currents. State cooling need not equal Hamiltonian projection. Conditional event samples need not describe the complete instrument. A stationary coefficient average need not equal correlated dynamical evolution. A Gram matrix must be whitened with the vertex. Orientation reversal need not be physical time reversal. Common rescalings cannot repair wrong ratios. Mixed-scale quark coordinates and running electroweak parameters cannot be compared directly with physical poles. A normwise residual cannot certify arbitrarily small singular values. Target-informed architecture or discrete selection remains target-informed even when runtime inputs are segregated. These distinctions delimit the claims of every calculation above.

The finite gauge/matter maps are useful structural constraints on this problem. They permit calibrated Standard Model charged-matter structure and exact tests of proposed repair coefficients, while leaving the physical source dynamics undetermined. The tested uncalibrated laws fail; their exclusions identify restricted classes rather than a contradiction of every OPH construction. The representation results, calibrated controls and physical-spectrum derivation therefore have separate conclusions.

## Reproduction

The source operators, invariant projectors and Higgs tensors are implemented in `model/observer_particles/`. The complete memory, gate and detector replay is `model/common_source/`

native_family_protocol.py; the chiral-candidate, hierarchy-diagnosis and full-repair-response suites contain the associated finite controls and exclusions. The working equations, acceptance conditions and evidence index are in docs/research/OPH_PARTICLE_MASS_RESPONSE.md and code/particles/PARTICLE_PROVENANCE_LEDGER.md. These executable receipts distinguish exact algebra, floating-point diagnostics and empirical comparisons. A hash check verifies custody; it does not replace execution or extend a theorem's physical scope.

# A Conditional modes and threshold kinematics

Declared Maxwell, flat-background pure Yang–Mills and two-derivative Minkowski Einstein–Hilbert actions have respectively two transverse, $2\dim G$ perturbative transverse, and two transverse-traceless classical modes. A quantum particle additionally needs a positive-energy physical Hilbert space and observable positive-residue spectral response. In particular, confined QCD is not assigned asymptotic free gluons by its perturbative mode count.

The complete thirty-seam source maps to $D_6 = \{z \in \mathbb{Z}^6 : \sum_i z_i \equiv 0 \pmod 2\}$, with a three-dimensional response-metric completion [6]. Proper-carrier naturality and a unique positive normalized minimizer give equal weights on sixty directed moves. Its spatial symbol has a two-dimensional momentum-orthogonal fiber at nonzero momentum. If that symbol is declared an oscillator stiffness under a nonzero coordinate dilation, $A_T'' + \Lambda_a(k)A_T = 0$, with $\omega_a^2 = \Lambda_a$, $\omega_a(0) = 0$. This supplies neither Gauss law nor a physical electromagnetic action, position or clock.

Its positive cosine symbol has an auxiliary root satisfying $|\Omega(k) - \Omega(p)| \le |k - p|$ in the chosen carrier chart. A physical photon identification and Lorentz-invariant positive-mass leptons would exclude photon decay into a pair under additive energy-momentum conservation. Other dispersion tests require their separate lepton, interaction and detector hypotheses. For leading dimension-six corrections, the head-on collinear pair threshold depends at positron share $x \in (0, 1)$ on

$$\delta_{\gamma,2} - x^3\delta_{+,2} - (1 - x)^3\delta_{-,2}.$$

Equal sharing therefore reads one combination of three coefficients; its coefficient fiber is two-dimensional. A photon-only bound assumes an independent Lorentz-invariant-lepton branch.

In the universal subcase with common hard-particle coefficient $\delta = -d = -a^2/20$, and a leading Lorentz-invariant soft photon, put $u = x(1 - x) \in (0, 1/4]$. The required soft energy is

$$\epsilon_{\mathrm{req}}(k, u) = \frac{m_e^2}{4ku} + \frac{3dk^3}{4}u.$$

The constrained minimum is at equal sharing when $3dk^4 \le 16m_e^2$; above that transition it is attained at $u_* = m_e/(\sqrt{3d}\,k^2)$, with $\epsilon_{\mathrm{env}} = (\sqrt{3}/2)m_e k\sqrt{d}$. Differentiation gives the minimizer; AM–GM gives the same envelope and excludes solutions below it. The executable calculation and the finite theorem verify the open-share equality witness and endpoint regime. These are conditional kinematic windows. They do not supply opacity, propagated flux, an air-shower interpretation or a physical attachment of $a$.

# B Finite source constraints on flavor

The icosahedral source constrains the carrier and repair dynamics before a mass readout is chosen. Let $N_{i+}, N_{i-}$ be the record counts at the two ports of antipodal pair $i$, and center their totals $b_i = N_{i+} + N_{i-}$ to obtain

$$W_5 = \{b \in \mathbb{R}^6 : \textstyle\sum_i b_i = 0\}.$$

For unit vectors $n_i$ along the six axes, define

$$T(b) = \sum_{i=1}^{6} b_i (n_i n_i^{\mathsf{T}} - I_3/3).$$

The icosahedral inner products give $T^*T = (4/5)P_5$, where $P_5 = I_6 - \mathbf{1}\mathbf{1}^{\mathsf{T}}/6$. Thus $W_5$ is the geometric quadrupole carrier. Its $A_5$ character has norm one, so it is irreducible; the invariant normalized linear functional on its endomorphisms is $\mathrm{Tr}\,/5$.

**A geometric response carrier and its heat channels.** Supply one unoriented face axis. Its normalizer is $D_3$, and exact restriction gives

$$W_5|_{D_3} = \mathbf{1} \oplus \mathbf{2} \oplus \mathbf{2}.$$

The group average is a canonical rank-one projector $P_0$ with matrix entries $\pm 1/6$ in pair coordinates. Its normalized trace is $1/5$; the complementary response has dimension four and trace weight $4/5$. This realizes the five-dimensional relabeling carrier geometrically. Selecting the face-axis background and identifying this quadrupole response with physical flavor remain additional steps. In particular, the physical geometric triplet restricts as $\mathbf{3}|_{D_3} = \mathrm{sign} \oplus \mathbf{2}$, a different representation.

There is no need to insert an arbitrary regular submodule to represent classical $D_3$ heat probabilities as channels on $W_5$. Conjugation gives

$$\mathrm{End}(W_5)|_{D_3} = 5\mathbf{1} \oplus 4\,\mathrm{sign} \oplus 8\mathbf{2}.$$

The Hilbert–Schmidt Gram matrix of its six group-conjugation superoperators is $24I_6 + J_6$, which is positive definite. Their mixture coefficients are therefore identifiable. If each of the three involutions has jump rate $c_2 \geq 0$ and each of the two three-cycles has rate $c_3 \geq 0$, the nontrivial eigenvalues of the positive heat generator are $6c_2$ and $3(c_2 + c_3)$. At time $t \geq 0$, put

$$r = e^{-3(c_2+c_3)t}, \qquad s = e^{-6c_2 t}; \qquad r^2 \leq s \leq 1.$$

The transposition-only relation $s = r^2$ requires $c_3 = 0$ for $t > 0$. The endomorphism heat trace is $5+16r+4s$; the regular-carrier trace $1+4r+s$ is a different observable. The full native 30-seam port Laplacian acts on $W_5$ as $6I_5$. A $D_3$ jump process is consequently a further dynamical hypothesis, not the native seam process expressed in another basis.

**A typed color coefficient.** Normalize fundamental color generators by $\mathrm{Tr}(T_a T_b) = \delta_{ab}/2$. Then $\sum_a T_a^2 = (4/3)I_3$, and the quadratic color observable

$$A = \Big(\sum_a T_a^2\Big) \otimes P_0 \quad \text{on } \mathbb{C}^3 \otimes W_5$$

has

$$\frac{\mathrm{Tr}\,A}{15} = \frac{4}{15}, \qquad \frac{\mathrm{Tr}\,A^2}{15} = \frac{16}{45}, \qquad \frac{\mathrm{Tr}\,e^{wA}}{15} = \frac{4 + e^{4w/3}}{5}.$$

This supplies a source realization of the linear coefficient $4/15$. The first four scalar cumulants are $4/15, 64/225, 256/1125, 1024/50625$; higher response terms are constrained once this particular observable and trace response are chosen. The observable is not a projector or a POVM effect: its nonzero eigenvalue is $4/3$. Its identification as a mass response is a physical hypothesis, and these cumulants do not derive a Gaussian law on arbitrary composite modules.

The contraction order matters. A color-invariant sharp projector on $\mathbb{C}^3 \otimes M$ has rank divisible by three. Twirling a rank-one component projector instead produces the unsharp invariant effect $(I_3/3) \otimes |m\rangle\langle m|$. It preserves the trace but divides the Hilbert–Schmidt square by three. Applying an isotropic Gaussian quadratic response after this twirl changes its coefficient accordingly; averaging nonlinear component responses afterwards need not do so. Neither source symmetry nor dimension counting fixes that choice. The exact character, channel, Casimir and moment identities are independently reconstructed by `code/particles/flavor/verify_source_w5_response_constraints.py`.

**Exact constraints from strict-descent seam repairs.** On a seam $(i, j)$, the strict-descent settlement rule transfers one record from $i$ to $j$ only if $N_i - N_j \geq 2$. It preserves $S = \sum_i N_i$ and decreases $V = \sum_i N_i^2$ by $2(N_i - N_j - 1) > 0$. Every signed-integer fiber $\{N : \sum_i N_i = S,\ V \leq B\}$ is finite and forward invariant. With the additional assumption of state-independent uniform proposals among the 30 seams, order the states by $V$. The transition matrix is triangular with diagonal

$$K_{NN} = 1 - \frac{a(N)}{30},$$

where $a(N)$ is the number of admissible seams. Its eigenvalues are exactly these diagonal entries, and every stationary measure is supported on settled states. The positive eigenvalues of $I - K$ have ratios at most 30. This is a discrete proposal-count statement; neither a physical clock nor equality between these rates and particle masses is assumed. The settlement rule differs from the native seam rule of the executable observer construction, which sets the two endpoint loads to the floor and ceiling of their mean and assigns an odd remainder by a fair coin. At load difference one the native rule waits or swaps, so its balanced class keeps moving; the triangular spectrum and the matching calculation below apply to the settlement rule only [11, 12].

A complete finite calculation is possible in the invariant sector

$$N_i \in \{0, 1, 2\}, \qquad \sum_{i=1}^{12} N_i = 12.$$

An active seam joins a port with count two to a port with count zero. Repair sets both counts to one and deletes both endpoints from the active bipartite graph; it creates no new active edge. Complete repair schedules are therefore precisely orderings of maximal matchings of that graph. This proves that the minimum settling cost is the smallest maximal-matching size, and the longest accepted schedule has the maximum-matching size. Uniform fresh seam proposals give a random-greedy-matching absorption law, without a Gaussian or diffusion approximation.

Exhaustive enumeration gives 73,789 states, 1,275 rotational orbits, and 303 settled states in nine orbits. Independent enumeration of 13,133 maximal matchings over orbit representatives verifies the settling calculation; the largest minimum cost is four, and the longest accepted schedule has six moves. The nine settled orbits leave an eight-dimensional simplex of $A_5$-invariant stationary mixtures. Symmetry and settling alone do not choose its weights.

The endpoint classification has a larger domain than that transient sector. Settledness implies $|N_i - N_j| \leq d(i, j)$ for graph distance $d$. Every icosahedral vertex has distance profile $(1, 5, 5, 1)$, hence $\sum_j d(i, j) = 18$ and

$$\frac{S}{12} - \frac{3}{2} \leq N_i \leq \frac{S}{12} + \frac{3}{2}.$$

If $S = 12k$ with integer $k$, integrality forces $N_i \in \{k-1, k, k+1\}$. Uniform shifts do not change the quadrupole. Consequently the 303 settled states and their four spectra exhaust *all signed-integer*

*endpoints* at such a total, without an initial three-level or positivity assumption. Strict descent makes every maximal accepted repair sequence finite; fresh uniform proposals reach an endpoint almost surely, whereas an unfair proposal schedule can stall. The settled quadrupole eigenvalue multisets are exactly

$$\{0,0,0\},\quad \{-2/\sqrt{5},0,2/\sqrt{5}\},\quad \{-1-1/\sqrt{5},\,1-1/\sqrt{5},\,2/\sqrt{5}\},$$

and the negative of the last multiset. Their multiplicities among settled states are respectively $63, 180, 30, 30$. Thus nonzero quadrupoles satisfy

$$\operatorname{Tr} Q^2 \in \{8/5, 16/5\}, \qquad \frac{(\operatorname{Tr} Q^3)^2}{(\operatorname{Tr} Q^2)^3} \in \{0, 9/64\}.$$

Under the additional affine log-mass readout $\log m_j = a + bq_j$, a nondegenerate ordered spectrum must consequently obey

$$\frac{\log(m_2/m_1)}{\log(m_3/m_2)} \in \{1, R, R^{-1}\}, \qquad R = \frac{3/\sqrt{5}-1}{2}.$$

This constrains a specified pointwise readout at every total divisible by twelve, but does not select the physical total, endpoint or vacuum. It does not apply the same finite menu to an arbitrary averaged response operator of a stationary mixture.

The unrestricted total also admits a finite classification. Quotient settled counts by uniform integer shifts and set their minimum to zero. The graph diameter bounds their range by three. Range-three extrema must be antipodal; the identity $d(a,j) + d(-a,j) = 3$ forces the unique height profile $N_j = 3 - d(a,j)$. Its antipodal pair totals are constant, so its quadrupole vanishes. There are twelve such profiles. Every other representative has entries in $\{0,1,2\}$ and minimum zero. Exhausting those assignments gives 6,065 settled representatives. Thus *all* signed-integer settled source states, at arbitrary total, reduce to 6,077 classes modulo uniform shift and exactly 31 quadrupole spectra. Their full characteristic polynomials and total residues modulo twelve are recorded in the executable certificate.

There is also a sharp norm bound on every integer settled state. The two seams between any two antipodal pairs form a perfect matching. The settled seam bound therefore gives $|b_i - b_j| \le 2$ for pair totals. The variance bound for six numbers in an interval of width two yields $\sum_i (b_i - \bar{b})^2 \le 6$. Hence

$$\operatorname{Tr} Q^2 \le \frac{24}{5}.$$

Equality is attained by setting all ports of three antipodal pairs to one and all ports of the other three pairs to zero. The bound extends to real assignments satisfying the same unit edge-Lipschitz condition; the integer repair threshold alone does not impose that condition on arbitrary real counts.

For every nonzero endpoint quadrupole, the shape invariant $\chi = (\operatorname{Tr} Q^3)^2/(\operatorname{Tr} Q^2)^3$ belongs to the following exact set of eleven values:

$$0,\quad \frac{5}{192},\quad \frac{45}{686},\quad \frac{9}{64},\quad \frac{1}{6},$$
$$\frac{487 \pm 189\sqrt{5}}{6000},\quad \frac{809 \pm 216\sqrt{5}}{29478},\quad \frac{12509 \pm 2808\sqrt{5}}{146334}.$$

The value $1/6$ means a repeated eigenvalue and cannot yield three distinct masses under the affine log readout. For its ordered adjacent log-gap ratio $g > 0$, centering a triple with gaps $1, g$ gives

$$\chi(g) = \frac{(g-1)^2(g+2)^2(2g+1)^2}{24(g^2+g+1)^3}.$$

This function is invariant under $g \mapsto 1/g$ and strictly decreases from $1/6$ to zero on $0 < g \leq 1$. The ten admissible values therefore give exactly nineteen possible log-gap ratios: one at $g = 1$ and nine reciprocal pairs. This finite restriction requires neither a chosen source total nor a noise approximation. It continues to require the stated individual-state mass readout; additive log scale, overall response strength, up/down attachment and physical mass scheme are not selected by the stationary-shape theorem.

For the actual accepted pair-total increment, rejection or deletion of an event can only decrease the sum of its outer products. Since each edge of the six-pair complete graph has two native seam representatives, uniform proposals give the statewise bound

$$0 \preceq \operatorname{Cov}(\Delta b \mid N) \preceq \mathbb{E}[\Delta b \Delta b^{\mathsf{T}} \mid N] \preceq \frac{2}{5} P_5.$$

On quadrupoles the covariance bound is $(8/25)I_5$ in Frobenius coordinates. At every settled state the accepted covariance vanishes. A symmetric proposal alphabet, Gaussian noise or a physical Poisson clock must not be inferred from this bound. The finite enumeration, matching identities and exact moment checks are reproduced by `code/particles/flavor/verify_native_repair_flavor_constraints.py`.

**A conditional driven record state.** A definite probability law for the relative record state follows from an additional preparation protocol using the source's existing atomic events. After each complete settlement, choose one of the twenty-four signed port events $(i, s)$, $s = \pm 1$, uniformly, append that event, and settle again using fresh uniform seam proposals. The choice of this renewal protocol is a hypothesis; it is not a consequence of the source grammar merely admitting the events. No mass target, fitted orbit weight, or physical time scale enters.

If $N$ is settled, the state during repair after this event has the form $N + se_j$. A repair is possible exactly along a seam from $j$ to $k$ with $s(N_j - N_k) = 1$, and moves the single defect from $j$ to $k$. Thus a particle walks strictly downhill in the original integer heights, and a hole strictly uphill. The graph diameter is three, so at most three repairs occur. Every accepted move has actual source–sink difference two. Rejected proposals change waiting counts but not the uniform choice among admissible neighbors.

Normalize heights by subtracting their minimum. On the resulting $6,077$ settled relative configurations the protocol defines a finite Markov kernel with $66,192$ nonzero entries. Removing a record at a global maximum leaves the state settled. Repeating reaches the zero representative, and reversing these direct events gives positive-probability paths back. The kernel is therefore irreducible. Every renewal flips the parity of the total residue, and each direct edge has a two-cycle, so its period is exactly two. There is a unique positive stationary distribution and Cesàro convergence; ordinary stepwise convergence from one parity class is not claimed.

Proper rotations give 136 orbits and a strongly lumped kernel with $1,000$ nonzero entries. The unique orbit weights are computed as exact rational numbers and independently certified on every one of the $6,077$ states. Their total-residue marginal is $1/12$ on each residue modulo twelve. The law also respects sign reversal. For the centered count quadrupole,

$$Q(N) = \sum_i N_i \left( n_i n_i^{\mathsf{T}} - \tfrac{1}{3} I \right), \qquad \mathbb{E}_\pi Q = 0, \qquad \mathbb{E}_\pi[Q \otimes Q] = \frac{\mathbb{E}_\pi \operatorname{tr} Q^2}{5} I_5.$$

The independently checked moments are

$$\mathbb{E}_\pi \operatorname{tr} Q^2 \simeq 1.4647638612, \qquad \mathbb{E}_\pi (\operatorname{tr} Q^2)^2 \simeq 2.8795832702.$$

All odd centered-count moments vanish. In particular, $\mathbb{E}_\pi \operatorname{tr} Q^3 = 0$. The radial fourth-moment difference from a centered isotropic five-dimensional Gaussian is

$$\mathbb{E}_\pi(\operatorname{tr} Q^2)^2 - \frac{7}{5}\big(\mathbb{E}_\pi \operatorname{tr} Q^2\big)^2 \simeq -0.1241631665 \neq 0.$$

These are moments of the declared record law; they cannot be transferred to an independently postulated physical response without its identification.

There is also an exact cost identity. Write $W(N) = \sum_i (N_i - S/12)^2$. A uniform signed preparation increases $W$ by $11/12$ on average. Each accepted one-defect repair decreases it by two. Since $W$ is bounded on the relative state space, stationarity gives

$$\boxed{\mathbb{E}_\pi[\text{accepted repairs per preparation}] = \frac{11}{24}.}$$

The stationary process is not reversible. For example, the transition from $e_{11}$ to $e_{10} + e_{11}$ has probability $1/20$, its reverse has probability $1/6$, and the exact stationary edge flows are unequal. This is a driven stationary record law, not an identified thermal equilibrium or quantum vacuum.

Only shift-invariant observables have this stationary law. The unquotiented protected total performs an unbiased walk on $\mathbb{Z}$ and has no normalizable stationary distribution. Subtracting a common height is a readout projection, not a physical gauge identification of different total records. Any Higgs or mass normalization depending on the mean count requires that additional physical input. Likewise, averaging an $A_5$-equivariant $\operatorname{End}(P_3)$ readout gives a scalar by irreducibility, even for a nonlinear readout. The distribution of individual sampled spectra is not the spectrum of that averaged operator. The renewal law fixes the orbit weights conditionally; it supplies no quark–Higgs map or mass scheme by itself.

**The full exterior Higgs response and its kinetic normalization.** Let the declared matter carrier be $C \oplus W = \mathbb{C}^3 \oplus \mathbb{C}^2$, with orthonormal exterior monomials, and write $H = h_0 e_3 + h_1 e_4$. The normalized top-form contractions

$$B_u(Q, H, u^c) = [Q \wedge H \wedge u^c]_5, \qquad B_d(Q, \overline{H}, d^c) = [(\iota_{\overline{H}} Q) \wedge d^c]_5$$

give, in the ordered color and weak basis,

$$U(H) = I_C \otimes \begin{pmatrix} h_1 \\ -h_0 \end{pmatrix}, \qquad D(H) = I_C \otimes \begin{pmatrix} -\overline{h}_0 \\ -\overline{h}_1 \end{pmatrix}.$$

Here the sign convention for $d^c$ can be absorbed into its independently supplied channel coefficient. Exact exterior and independent creation-operator calculations give

$$U^\dagger U = D^\dagger D = \|H\|^2 I_C, \qquad U^\dagger D = 0.$$

Thus the full doublet, including its charged direction, has unit-magnitude color Clebsches. The channel coefficients $g_u, g_d$ are not determined by normalizing these invariant tensors. Color covariance forces the color intertwiner to be a scalar identity; the traceless three- and five-dimensional parts of the internal color current cannot be read as family masses.

On the conditional family attachment $F \otimes M$, restricting every supplied operator to $I_F \otimes A$ gives $Y_u = g_u I_F$ and $Y_d = g_d I_F$. This restriction is preserved by products, adjoints, inverses, positive functional calculus, and elimination of heavy modes with a family-independent block decomposition. It is a precisely bounded direct-contraction model, not a classification of every source interaction.

It produces no hierarchy. After converting the all-left Weyl coefficients to the four-component Dirac Yukawa convention, a local chiral action with positive, Higgs-independent kinetic matrices has normalized vertex

$$Y_{q,\mathrm{can}} = Z_Q^{-1/2} Y_q Z_q^{-1/2} / \sqrt{Z_H}.$$

Here $Z_q$ is the right-handed kinetic metric. For a literal all-left bilinear the left normalization factor instead carries a transpose; the two coefficient conventions must not be mixed. Consequently a proposed record matrix $R > 0$ with $Z_Q = Z_q = R$ and $Y_q = g_q R$ cancels from the vertex: $Y_{q,\mathrm{can}} = g_q I_F / \sqrt{Z_H}$. Equivalently, $\det(pR - h g_q R) = \det(R)(p - h g_q)^3$. This is a conditional identity; no source selection of $Y_q = g_q R$ is assumed. An interacting source completion must therefore emit a Higgs derivative that is distinguishable from its kinetic metric, together with the common state and normalization. For a differentiable quadratic kernel with blocks $A, B, C, D$, that derivative after eliminating an invertible heavy block is

$$\partial_H (A - BD^{-1}C) = A_H - B_H D^{-1} C - B D^{-1} C_H + B D^{-1} D_H D^{-1} C.$$

The finite carrier construction does not supply this Higgs-dependent source kernel or a physical mass prescription.

The independent finite replays are `code/particles/flavor/verify_native_record_renewal.py` and `code/particles/flavor/verify_exterior_higgs_response.py`.

### B.1 Stabilizers and entropy-selected quadrupoles

Residual symmetry alone does not select the mass ratios. Realize $W_5$ as the traceless symmetric real $3 \times 3$ matrices. Invariance under a threefold or fivefold rotation about $n$ forces

$$A = \alpha(nn^T - I/3),$$

with spectrum $(2\alpha/3, -\alpha/3, -\alpha/3)$. Those fixed points have a double eigenvalue. The twofold fixed locus is

$$A = \begin{pmatrix} u & v & 0 \\ v & w & 0 \\ 0 & 0 & -u - w \end{pmatrix},$$

which is three-dimensional, hence two-dimensional after overall scaling, and admits simple spectrum. It leaves exactly enough continuous freedom to carry the two independent mass ratios without predicting them. The Klein four-group of three perpendicular twofold axes fixes the diagonal traceless forms, one projective parameter with generic simple spectrum, and every subgroup of $A_5$ other than the trivial group, $C_2$, and $V_4$ contains a threefold or fivefold rotation, so no residual stabilizer pins an isolated simple-spectrum orbit. The source-side family problem is therefore selection by a specific screen-derived $A_5$-invariant potential.

The leading cubic entropy truncation has a global classification. A $W_5$ port vector consists of six values $a_i$, each repeated on an antipodal pair, with $\sum_i a_i = 0$ and $2\sum_i a_i^2 = 1$. At a constrained extremum of $S_3 = 2\sum_i a_i^3$, the Lagrange equation makes every $a_i$ a root of one quadratic. Hence there are at most two distinct values, and enumerating their multiplicity $m = 1, \ldots, 5$ gives

$$S_{3,m} = \frac{3 - m}{\sqrt{3m(6 - m)}}.$$

The global maximum is $2/\sqrt{15}$, uniquely up to permutation, with one antipodal pair against the other five. It is the $C_5$-axis orbit, and its quadrupole spectrum is exactly $(-2/\sqrt{15}, -2/\sqrt{15}, 4/\sqrt{15})$.

Thus the leading cubic truncation selects a double eigenvalue and fails the required simple-spectrum shape. Interpreting that degeneracy as two equal physical charged masses requires the separately unproved $W_5$-to-family attachment (`code/particles/runs/leptons/charged_entropic_branch_no_go.json`).

At finite band amplitude, the global minimizers of one coefficient-free candidate—the quartic truncation of the relative-entropy expansion on $W_5$—can also be classified exactly. Write $b_i = 1 + ra_i = 12q_i$. Exact stationarity, second-variation, and face factorizations show that every global interior minimizer has two coordinate values and that no omitted face wins. With

$$r_c = \frac{32\sqrt{15} - 20\sqrt{6}}{27} = 2.7757656378\ldots,$$

the strict record domain $b_i > 0$ selects the degenerate multiplicity-one $C_5$ orbit for $0 < r < r_c$, both multiplicity-one and multiplicity-two orbits at $r = r_c$, and the simple-spectrum multiplicity-two orbit for $r_c < r < \sqrt{24}$. The latter has centered eigenvalues proportional to $(-(\varphi^4 - 1), -1, \varphi^4)$ and sorted-gap ratio $\varphi$. For $\sqrt{24} \le r < \sqrt{60}$, the strict problem has no attained minimum; its infimum lies on the boundary.

The quartic truncation also admits a simple-spectrum boundary branch. On the closed probability simplex the global minimizer above $\sqrt{24}$ has four pair weights zero and two weights $b_\pm = 3 \pm \frac{1}{2}\sqrt{r^2 - 24}$. Its quadrupole spectrum is proportional to $(-2, 1-d, 1+d)$, where $d = \sqrt{(r^2 - 15)/5}$, so its upper-to-lower sorted-gap ratio is

$$R(r) = \frac{2d}{3 - d}, \qquad r^2 = 15 + \frac{45R^2}{(R+2)^2}.$$

It reaches the compare-only central charged-lepton ratio $R = 1.889009\ldots$ at the target-attached value $r = 5.061326\ldots$. This is not a prediction: eight of twelve port probabilities then vanish, the Taylor expansion about the uniform record is least controlled there, no source emits $r$, and the physical quadrupole-to-log-mass attachment is not supplied. The strict full-support quartic minima are conditionally excluded as the observed shape. The closed boundary remains a minimum of that truncated objective (`code/particles/runs/flavor/entropy_w5_shape_certificate.json`).

**Full relative entropy at fixed pair norm.** The untruncated uniform-reference objective gives a sharper answer for the same six-pair model. Set

$$p_i = \frac{1 + ra_i}{6}, \qquad \sum_i p_i = 1, \qquad \sum_i p_i^2 = \frac{1}{6} + \frac{r^2}{72}, \qquad p_i \ge 0,$$

and minimize $D(p\|\mathbf{1}/6) = \sum_i p_i \log(6p_i)$, with $0 \log 0 = 0$. Pairing the equal antipodal port probabilities leaves this divergence unchanged. For every $0 < r \le \sqrt{60}$, all global minima are the six permutations of

$$p_{\text{high}} = \frac{1 + 5r/\sqrt{60}}{6}, \qquad p_{\text{low}} = \frac{1 - r/\sqrt{60}}{6} \quad \text{(five entries)}.$$

This is the standard maximum-Shannon-entropy branch at fixed index of coincidence [18]; its implication for the icosahedral quadrupole is the application here.

For completeness, compactness gives a minimum. At an interior extremum with $0 < r < \sqrt{60}$, the two constraint gradients are independent, and stationarity gives $\log p_i + 1 = \lambda + 2\mu p_i$. Strict concavity of $\log x - 2\mu x$ allows at most two positive roots $x < y$. If the larger root occurs twice,

the variation exchanging those two coordinates is tangent to both constraints, while its constrained Hessian coefficient is negative:

$$\frac{1}{y} - 2\mu = \frac{1}{y} - \frac{\log y - \log x}{y - x} < 0.$$

Thus an interior minimum has one high and five low entries, and the two constraints give the displayed values.

No nonvertex boundary point minimizes the full objective. If its positive coordinates include two unequal values, adding a new weight $\epsilon > 0$ at a zero coordinate and adjusting those two values by $O(\epsilon)$ preserves both constraints, by the implicit function theorem. The change in the objective is $\epsilon \log \epsilon + O(\epsilon) < 0$. If instead the support consists of $k \geq 2$ equal weights $1/k$, replace those weights by $(1 - \epsilon)/k$, add the new weight $\epsilon$, and add and subtract $\delta$ at two old coordinates, where

$$\delta^2 = \frac{\epsilon}{k} - \frac{k+1}{2k}\epsilon^2.$$

The perturbed weights stay positive for sufficiently small $\epsilon$, both constraints hold exactly, and the old-coordinate contribution changes by $O(\epsilon)$; the same negative leading term excludes this boundary point. At $r = \sqrt{60}$, purity is one and only the six vertices are feasible. At $r = 0$, the distribution is uniform and does not select a normalized direction $a$.

For $r > 0$, every selected normalized port direction has pair values $(5, -1, -1, -1, -1, -1)/\sqrt{60}$, up to permutation. With six unit antipodal-axis representatives $n_i$, the forward twelve-port map is $Q(a) = 2\sum_i a_i(n_i n_i^{\mathsf{T}} - I/3)$; its normalization obeys $n_i^{\mathsf{T}} Q(a) n_i = 8a_i/5$. Its spectrum is

$$\operatorname{spec} Q(a) = \left(-\frac{2}{\sqrt{15}}, -\frac{2}{\sqrt{15}}, \frac{4}{\sqrt{15}}\right).$$

The six branches are the antipodal vertex axes; each has the order-ten dihedral stabilizer $D_5$ in the rotational icosahedral group (the $C_5$-axis terminology names its fivefold axis). The perturbation quadrupole $Q(ra) = rQ(a)$ is zero at $r = 0$ and has a repeated eigenvalue at every positive feasible amplitude. Consequently, no amplitude rule $r = r(P)$ rescues three distinct family masses through an attachment that reads only these quadrupole eigenvalues and treats equal eigenvalues equally. This excludes that uniform-reference, fixed-norm full-entropy selector with that readout, including its closed boundary; it does not exclude OPH, other references or constraints, different actions, or different physical readouts. The quartic boundary formula remains valid for its own objective; its two-positive-weight branch does not minimize the full entropy, although both objectives share the pure endpoint. Neither a physical mass attachment nor numerical quark masses follow from this result (`code/particles/runs/flavor/full_entropy_w5_selector.json`).

The finite theorem does not supply an effective action on $W_5$ with a unique simple-spectrum minimizing orbit. The invariant algebra has one quadratic, two cubic, and two quartic coefficients, so the $W_5$-restricted Hessian scalar $h_5$ at the homogeneous state is a necessary local discriminator. A physical charged shape additionally requires the physical $A_5$ family lift, normed determinant-line descent with kinetic factors, an interacting charged kernel, an operational scale, and a source packet fixed before comparison.

# C Perturbative intrinsic neutrino response

For the isotropic matrix put $\varphi = \Omega/3$, $m_d^2 = |a - \rho e^{i\varphi}|^2$, and $\Delta_{\mathrm{atm,iso}} = |a + 2\rho e^{i\varphi}|^2 - m_d^2$. The following expansion uses the branch on which the collective state is above the doublet. Write

$$\psi_e = \varphi + \delta_e, \qquad \delta_{12} + \delta_{23} + \delta_{31} = 0.$$

At first order in the centered eta-class,

$$\delta_e = -\tan\varphi\,\eta_e + O(\eta^2).$$

Define

$$\sigma^2 = \frac{2}{3}\sum_e \delta_e^2.$$

Assume $2a\cos\varphi + \rho \neq 0$ and that the collective state is separated from the isotropic doublet. Then the two ascending doublet eigenvalues satisfy

$$s_{0,1}^2 = m_d^2 \mp 2a\rho|\sin\varphi|\,\sigma + O(\delta^2),$$

so

$$s_1^2 - s_0^2 = 4a\rho|\sin\varphi|\,\sigma + O(\delta^2) = 4a\rho\frac{\sin^2\varphi}{|\cos\varphi|}\sqrt{\frac{2}{3}\sum_e \eta_e^2} + O(\eta^2).$$

Under the declared normal-ordering hypothesis, the ascending atmospheric gaps obey

$$\Delta m_{31}^2 = \Delta_{\mathrm{atm,iso}} + \frac{1}{2}\Delta m_{21}^2 + O(\delta^2), \qquad \Delta m_{32}^2 = \Delta_{\mathrm{atm,iso}} - \frac{1}{2}\Delta m_{21}^2 + O(\delta^2),$$

so the first-order invariant atmospheric object is the collective-to-doublet centroid gap

$$\Delta_{\mathrm{cent}} = s_2^2 - \frac{1}{2}(s_0^2 + s_1^2) = \Delta_{\mathrm{atm,iso}} + O(\delta^2).$$

Its quadratic shift is

$$\delta\Delta_{\mathrm{cent}} = -a\rho\,\sigma^2\frac{a(4\cos^2\varphi - 1) + 6\rho\cos\varphi}{2a\cos\varphi + \rho} + O(\delta^3).$$

The projective largest-mass right-singular direction, in the phase gauge that approaches the real democratic vector at the isotropic point, obeys the first-order deformation law

$$u_3 = u + \kappa\begin{pmatrix}\delta_{23}\\ \delta_{31}\\ \delta_{12}\end{pmatrix} + O(\delta^2), \qquad u = \frac{1}{\sqrt{3}}(1,1,1)^T, \qquad \kappa = \frac{\sqrt{3}\,(2a\sin\varphi + 3i\rho)}{9(2a\cos\varphi + \rho)}.$$

This equation fixes only the complex line. If $v_3$ denotes its normalized right-hand side, the canonical Takagi representative used by the executable construction is

$$u_3^{\mathrm{Tak}} = e^{-\frac{i}{2}\arg(v_3^T M_\nu v_3)}v_3, \qquad (u_3^{\mathrm{Tak}})^T M_\nu u_3^{\mathrm{Tak}} = s_3 > 0,$$

up to the unavoidable column sign. In particular, at the isotropic point the real democratic vector is generally outside positive Takagi gauge. Equivalently,

$$u_3 = u - \tan\varphi\,\kappa\begin{pmatrix}\eta_{23}\\ \eta_{31}\\ \eta_{12}\end{pmatrix} + O(\eta^2).$$

The expansion requires a separated collective state and $2a\cos\varphi + \rho \neq 0$. It concerns the conditional matrix above. A weighted-cycle comparison has additional unselected basis permutation, holonomy orientation, CP sign and absolute attachment. Enumerating both stored orientations and all row/column relabelings gives no normal-ordering-consistent matrix passing both adopted correlated oscillation gates. This excludes a convention-only repair of that matrix, without classifying source-derived charged-basis placements.

# D Affine event-record continuation

The following conditional construction separates event supports, observer frames, clocks and gauge transport. It is included to specify exactly what record continuation establishes before any particle or geodesic attachment.

The compact cap-normal theorem supplies the Lorentz-natural frame hyperboloid. It identifies sky directions with $q(\Omega) = (1, \Omega)$, represents oriented round caps by $n_C = (\cot\alpha, \csc\alpha\, \mathbf{c})$, and maps each cap to a geodesic half-space in $H^3$. The source-causal continuum certificate separately supplies affine physical event charts only after exact two-way cone/order agreement, count–volume, manifoldlikeness, topology, and refinement convergence. This section consumes event supports, observer clocks, chart transitions, gauge transport, and worldline stitching. Frame estimation, event location, overlap descent, and worldline stitching are separate certificates.

Fix a curvature radius $R_H > 0$ and write

$$H^3_{R_H} = \{X \in \mathbb{R}^{1,3} : \langle X, X\rangle_L = -R_H^2,\ X^0 > 0\}, \qquad \langle X, Y\rangle_L = -X^0Y^0 + X^1Y^1 + X^2Y^2 + X^3Y^3.$$

The hyperbolic distance used for conditioned frame comparison is

$$d_H(X, Y) = R_H \operatorname{arcosh}\left(-\frac{\langle X, Y\rangle_L}{R_H^2}\right).$$

An event-record atlas is a tuple

$$\mathcal{A}_H = (R_H, \{U_i, \chi_i, u_i, J_i, \vartheta_i\}, \{G_{ji}, F_{ji}\}, E, \nabla, \mathsf{Prov}_{R_H}, \mathsf{Prov}_u)\,.$$

Here $\chi_i$ are affine event charts, $u_i \in H^3_{R_H}$ is the selected local observer-frame anchor when such an anchor is claimed, $J_i$ are event-chart domains for record support, and $\vartheta_i : J_i \to T$ are observer-clock maps into a common comparison-time line $T$. $\mathsf{Prov}_{R_H}$ records whether $R_H$ is a unit convention, an imported physical scale, or an independently derived OPH scale. The unit-chart convention alone supplies no independent OPH scale. $\mathsf{Prov}_u$ records which observer, tetrad, or clock object selected the anchors $u_i$. The Lorentz part $G_{ji} \in \mathrm{SO}^+(1,3)$ transports frames and tetrads; the full Poincaré transition transports affine event coordinates. The map $F_{ji}$ is the sector or gauge transport functor on the record payload. These two holonomies are deliberately separate: atlas holonomy tests geometry, and gauge holonomy tests transported sector data. When two charts describe the same transported observer frame, the atlas additionally requires

$$G_{ji}u_i = u_j.$$

**Theorem D.1** (H3 geometric naturality)**.** *For every $A \in \mathrm{SO}^+(1,3)$, the map $X \mapsto AX$ preserves $H^3_{R_H}$, $d_H$, geodesics, exponential and logarithm maps, parallel transport, Hausdorff distances between compact frame supports, and Fréchet means whenever the mean is unique.*

*Proof.* The defining equation of $H^3_{R_H}$ and the formula for $d_H$ depend only on the Lorentz inner product and the future sheet. Elements of $\mathrm{SO}^+(1,3)$ preserve both. The Levi-Civita connection, geodesic exponential, logarithm, and parallel transport are functorial for isometries, so the listed constructions commute with $A$. Hausdorff distance and unique Fréchet means are metric constructions, hence are preserved by any isometry. □

## D.1 Cap-response frame balls and affine event supports

The Standard Model gauge paper supplies the conditional inverse theorem that turns calibrated modular cap responses into conditioned observer-frame supports. For every descended record token $i$ and clock slice $t$, the frame layer consumes a certificate

$$R_i(C,t,O) = \omega_{i,O}\Big(\sigma_t^{C,O}(M_{C,0,O})\Big), \qquad y_{i,j} = F_j(X_i(t)) + e_{i,j},$$

on a compact frame domain $\Omega \subset H^3_{R_H}$, with quantitative observability

$$\alpha\bar{d}(X,Y) \le |F(X) - F(Y)|_W \le L\bar{d}(X,Y), \qquad |e_i|_W \le \sigma_i(t).$$

Given an $\varepsilon$-net and residual tolerance $\tau_i(t)$, the imported theorem gives the conservative enclosure

$$S_i(t) \subseteq B_H(\widehat{X}_i(t), r_i(t)), \qquad r_i(t) = R_H\left[\frac{L}{\alpha}\varepsilon + \frac{2}{\alpha}\sigma_i(t) + \frac{1}{\alpha}\tau_i(t)\right].$$

Not every point in the displayed ball is asserted to fit the responses. A frame estimate $\widehat{X}_i(t)$ is a unique finite output only when $\Delta_{\mathrm{frame},i}(t) > 0$. If this gap fails, the output is a frame ambiguity set or support ball. This ball cannot serve as an event position. Event stitching consumes an affine event support $A_i(t)$ produced by the physical source-causal continuum and its ancestry/placement receipts. Distinct event supports at a shared clock slice require a certified separation in the affine event metric. A conditioned-frame separation may also be recorded:

$$d_H(\widehat{X}_i, \widehat{X}_k) > r_i + r_k + m_{\mathrm{sep}}$$

for a declared positive frame margin $m_{\mathrm{sep}}$. It does not replace event separation.

## D.2 Proto-record extraction and descent

A raw local proto-record is a connected component of an invariant detector scalar above a frozen threshold, together with its certified affine event support, conditioned frame ball, local clock interval, sector payload, and uncertainty collar. The detector scalar is built from record-algebra data, not from implementation object IDs. Extraction is chart-natural when transporting the raw field by $G_{ji}$ and $F_{ji}$ carries extracted components in chart $i$ to the extracted components in chart $j$, up to the declared support collar.

Before temporal stitching, overlapping charts are descended. The overlap-descent relation joins two proto-records only when their transported affine event supports lie in the same connected component on a real chart overlap and the same-component margin dominates the support collar. If distinct component separation does not dominate the collar, the descent emits an ambiguity certificate instead of choosing a label. On component-connected covers, this descent produces canonical global descended records by the usual sheaf gluing argument: cocycle-compatible local components with positive separation margins have a unique global component, and the construction is natural under atlas restriction.

## D.3 Boundary-crossing germs and transported residuals

A cross-boundary candidate edge is admissible only when the two descended records meet a real patch interface in adjacent observer-clock intervals and share a common comparison chart. The interface contact must be oriented and transverse: the signed-distance interval brackets the interface,

and the normal velocity has a positive lower bound. Tangencies, grazing contacts, and file or shard-name boundaries without a physical interface are rejected. If the local slab contains an interaction marker, the record is routed to the interaction solver instead of stitched by the free-propagation rule.

For an admissible pair, the certificate transports the left record into the right comparison chart and evaluates frozen residuals

$$r_t, \quad r_\partial, \quad r_x, \quad r_v, \quad r_{\mathrm{int}},$$

for clock adjacency, boundary contact, affine event position, transported velocity or finite difference, and interaction absence. The cost function is declared before seeing the proposed matching. Record IDs, global IDs, shard-local IDs, and stitch keys are forbidden in admissibility, costs, and tie-breaking. All affine event-position and velocity residuals are evaluated as certified interval or uncertainty-box distances. A point-estimate residual may be printed as a diagnostic; it cannot replace the event-support radius in an admissibility or matching certificate. Conditioned $H^3$ frame residuals are a separate tetrad-consistency check.

**Lemma D.2** (Candidate-edge naturality)**.** *The admissible candidate-edge set and its residual intervals are invariant under common* $\mathrm{SO}^+(1,3)$ *chart changes and under gauge-bundle trivialization changes.*

*Proof.* Every event term is computed from the affine overlap cocycle, oriented interface data, and the common clock line. Frame terms use $d_H$ and Theorem D.1. Sector and gauge residuals compare transported payloads after applying the declared connector, so changing a local trivialization conjugates both sides of the comparison. The residual intervals and the resulting admissibility predicate are therefore unchanged. □

## D.4 Assignment gap, nonbranching paths, and ambiguity

Let $L$ and $R$ be the descended record sets on two adjacent time slabs. A stitch matching is a partial one-to-one assignment between $L$ and $R$, augmented by declared appearance and disappearance interval costs. Its certified gap is

$$\Delta_{\mathrm{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*),$$

where $M_*$ is the proposed matching, $\overline{C}(M_*)$ is its upper cost bound, and $\underline{C}(M)$ is the lower cost bound for every competing matching. The event-location gap $\Delta_{\mathrm{event}}$ and stitch-assignment gap $\Delta_{\mathrm{cert}}$ are distinct receipts: the first certifies a unique event support inside one clock slice, while the second certifies a unique temporal assignment across slices.

**Theorem D.3** (Stable ID-independent stitch assignment)**.** *If the candidate-edge set is natural, the costs are frozen and ID-independent, the proposed matching is one-to-one, and* $\Delta_{\mathrm{cert}} > 0$*, then* $M_*$ *is the unique certified minimum-cost stitch. The certified matching is natural under chart changes, gauge trivializations, and relabeling of implementation record IDs.*

*Proof.* The strict gap says every competing matching has lower cost bound above the proposed matching's upper cost bound, so no competitor can tie or beat it under any realization inside the certified intervals. Lemma D.2 makes the candidate graph and cost intervals invariant under chart and gauge presentation changes. Because IDs are excluded from admissibility and costs, relabeling them changes only bookkeeping fields, not the optimum. □

**Theorem D.4** (Unavoidable ambiguity)**.** *If a nontrivial permutation of the visible descended records preserves the atlas, clocks, interfaces, sector/gauge payloads, candidate graph, and cost intervals, and its induced action fixes no admissible stitch, then no natural ID-independent finite procedure can certify a unique stitch across that slab. Without the fixed-point-free clause, a unique natural stitch need only be invariant under the symmetry and ambiguity does not follow.*

*Proof.* A natural procedure must commute with the permutation. A unique certified output would therefore be a fixed admissible stitch, contrary to the fixed-point-free hypothesis. Choosing one element of a nontrivial orbit would introduce a label or presentation dependence. Thus no natural single-valued admissible-stitch output depending only on the listed preserved visible inputs exists. The declared fail-closed protocol records this situation as AMBIGUOUS; an invariant orbit/set-valued output or rejection is a different protocol and is not excluded. □

The certified stitch graph is then the graph obtained by chaining the unique matchings across adjacent slabs. Since each matching is partial one-to-one, every connected component is a non-branching path, ray, or isolated record. These are the affine event-record worldlines of this section.

## D.5 Refinement naturality and dynamical lift

For a refinement $s \to r$, require that record contraction induces a map $Q_{sr} : \mathcal{M}_s \to \mathcal{M}_r$ from every admissible fine matching to an admissible coarse matching. The coarse optimum $M_r^*$ must have at least one admissible fine lift $\widetilde{M}_s$, with $Q_{sr}(\widetilde{M}_s) = M_r^*$. Suppose the certified cost bounds imply

$$|C_s(M) - C_r(Q_{sr}M)| \le \eta_{sr} \qquad (M \in \mathcal{M}_s)$$

for all admitted realizations, and the coarse certified gap obeys $\Delta_r > 2\eta_{sr}$. If a fine optimum $M_s^*$ contracted to a competitor, then

$$C_s(M_s^*) \ge C_r(M_r^*) + \Delta_r - \eta_{sr} > C_r(M_r^*) + \eta_{sr} \ge C_s(\widetilde{M}_s),$$

a contradiction. Thus every fine optimum contracts to $M_r^*$. The lift premise is essential: coarse costs $C_r(A) = 0, C_r(B) = 1$, a single fine matching mapping to $B$ with cost 1, and $\eta_{sr} = 0$ satisfy the gap and distortion inequalities but cannot recover $A$. This is the refinement-naturality gate: feasibility, the lift, and the uniform cost bound must all be certified before the coarse/fine agreement follows from the margin.

The stitch certificate lives on the conditional effective Lorentzian manifold. A geodesic or free-particle interpretation requires a separate dynamical receipt with a timelike speed margin, connection compatibility, and an action or propagation law. The stitch theorem proves token continuation in event charts; it does not prove geodesic motion.

**Theorem D.5** (Certified affine event-record stitching)**.** *Given a source-selected refinement family satisfying the spacetime paper's source-causal continuum certificate, together with a valid affine event atlas compatible with its effective limit, a common operational clock atlas, affine event supports for every descended record token and clock slice, and separately typed conditioned-frame certificates where observer frames are used, chart-natural proto-record extraction, overlap descent with positive event-support separation margins, real transverse interface-crossing germs, sector and gauge transport continuity, a frozen ID-independent one-to-one assignment with $\Delta_{\mathrm{cert}} > 0$, no interaction marker in the free-propagation slab, and a passing coarse/fine contraction gate, the emitted stitch graph is a unique natural nonbranching affine event-record graph. Under the declared fail-closed protocol, failure to certify a positive event-location gap, separation margin, or assignment gap returns* AMBIGUOUS *and certifies no unique stitch by this gap contract. That protocol output is not*

*a proof of mathematical nonuniqueness; unavoidable ambiguity additionally requires the fixed-point-free symmetry hypothesis of Theorem D.4. The protocol returns* INTERACTION REQUIRED *when an interaction marker is present, and rejects the stitch when its atlas, interface, event-location, or transport gates fail.*

*Proof.* The atlas, event-location, and extraction gates make local proto-records presentation-independent. The positive $\Delta_{\mathrm{event}}$ gate licenses event-position outputs; otherwise the procedure carries event boxes or ambiguity sets forward. Descent turns overlap-compatible local components into descended records before any temporal decision is made. The interface and transport gates define a natural candidate-edge graph by Lemma D.2. The strict assignment gap gives a unique ID-independent matching by Theorem D.3. Chaining partial one-to-one matchings gives nonbranching components, and the refinement gate makes the result stable under the declared coarse/fine contraction. The other labels are the declared protocol's fail-closed responses to failed or uncertified gates; they need not be disjoint or logically exhaustive. Theorem D.4 upgrades the ambiguity label to a mathematical obstruction only under its fixed-point-free action on admissible stitches. □

# E Heat-Kernel Calibration Supplement

This appendix gives the compact-group heat-kernel calculation used by the particle calibration. The fixed-cutoff same-overlap thermal/Casimir law is proved in Ref. [6]. The calculation below supplies the compact-group Peter–Weyl lift and beta-shift bookkeeping. It does not derive the physical one-loop beta coefficients from edge sectors alone.

## E.1 $\mathcal{R}_U$ Unified-Coupling Certificate

The $\mathcal{R}_U$ certificate supplies the unified diffusion coupling used by the hierarchy discussion of the flagship synthesis and by the electroweak transmutation factor

$$\frac{v}{E_\star} = P^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P)}\right].$$

It does not certify the full clock branch

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}_{\mathrm{QCD/nuc}}^{^{133}\mathrm{Cs}} + \mathcal{R}_{\mathrm{atom}}^{^{133}\mathrm{Cs}}.$$

The complete $\mathcal{R}_\gamma$ construction requires a source-only electromagnetic endpoint, charged-lepton absolute scale, cesium nuclear data, and atomic spectral enclosures. The numerical package is a witness for $\mathcal{R}_U$. Its mathematical conclusion requires the stated outward-rounded interval arithmetic.

**Declared source packet.** The public-pixel branch uses

$$P_C = \frac{\texttt{1.630968209403959324879279847782648941335982851627925060666 1}}{\texttt{507533907793398933432}}.$$

The source-only branch uses the certified coordinate

$$P_{\mathrm{fwd}} = 1.630972095858897\ldots.$$

It is interval-certified unique on its interval and, by the domain-global certificate, the only fixed point of its readout map on the declared analytic domain $\alpha^{-1} \in [100, 200]$ (the interval contraction

certificate of 2026-07-14, whose closure row is a certified source-root row). The finite heat-kernel cutoffs are

$$N_2 = 128, \qquad N_3 = 64.$$

The one-loop running packet is

$$(b_1, b_2, b_3) = \left(\frac{33}{5}, 1, -3\right), \qquad N_c = 3, \qquad \beta_{\mathrm{EW}} = N_c + 1 = 4.$$

This packet is an explicit premise of the certificate. This appendix does not derive it from the recovered OPH core.

**Definition E.1** (No-hidden-free-variable rule for $\mathcal{R}_U$)**.** *The following objects must be fixed before comparison with electroweak or gravity data:*

$$(b_1, b_2, b_3), \quad N_2, \quad N_3, \quad M_U(P), \quad E_{\mathrm{cell}}(P),$$
$$\beta_{\mathrm{EW}}, \quad \textit{Casimir normalization}, \quad \textit{matching scheme}.$$

*Changing any of them after comparison with $M_W$, $M_Z$, $\alpha_s(M_Z)$, low-energy electroweak couplings, or $G_{\mathrm{exp}}$ demotes the row to calibration.*

**Dimensionless source map.** Set $E_\star = 1$ for the dimensionless calculation. For a trial coupling $a$, define

$$M_U(P) = e^{-2\pi} P^{1/6}, \qquad E_{\mathrm{cell}}(P) = P^{-1/2},$$
$$v(P, a) = E_{\mathrm{cell}}(P) \exp\left[-\frac{2\pi}{4a}\right].$$

For $i = 1, 2, 3$,

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \qquad \alpha_Y(\mu; P, a) = \frac{3}{5}\alpha_1(\mu; P, a).$$

The source $Z$-scale satisfies the implicit equation

$$\mu = \frac{v(P, a)}{2} \sqrt{4\pi\alpha_2(\mu; P, a) + 4\pi\alpha_Y(\mu; P, a)}.$$

For SU(2), with $j = n/2$ and $0 \le n \le N_2$,

$$d_j = 2j + 1, \qquad C_2(j) = j(j + 1).$$

For SU(3), with $0 \le p, q \le N_3$,

$$d_{p,q} = \frac{(p + 1)(q + 1)(p + q + 2)}{2}, \qquad C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}.$$

The group sums are

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \qquad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

Define

$$t_2(P, a) = 4\pi^2 \alpha_2(\mu_Z(P, a); P, a), \qquad t_3(P, a) = 4\pi^2 \alpha_3(\mu_Z(P, a); P, a),$$

and

$$\Phi_U(P, a) = \bar{\ell}_{\mathrm{SU}(2)}(t_2(P, a)) + \bar{\ell}_{\mathrm{SU}(3)}(t_3(P, a)) - \frac{P}{4}.$$

The certificate value is the zero $\Phi_U(P, \alpha_U(P)) = 0$.

**Numerical witness.** For $P = P_C$, the checker emits

$$\alpha_U(P_C) = 0.041124336195630495,$$
$$\alpha_U(P_C)^{-1} = 24.316501918546496.$$

The corresponding dimensionless hierarchy data are

$$\frac{M_U(P_C)}{E_\star} = 0.0020260720012100037,$$
$$\frac{v(P_C)}{E_\star} = 2.0199803239725553 \times 10^{-17},$$
$$\frac{\mu_Z(P_C)}{E_\star} = 7.502403238986022 \times 10^{-18}.$$

The emitted couplings at $\mu_Z$ are

$$\alpha_1 = 0.016885708038988166,$$
$$\alpha_Y = 0.010131424823392899,$$
$$\alpha_2 = 0.033777889448908055,$$
$$\alpha_3 = 0.11833602747445048.$$

The heat-kernel parameters and entropy readouts are

$$t_2 = 1.3334976254578108,$$
$$t_3 = 4.6717191102770705,$$

$$\bar{\ell}_{\mathrm{SU}(2)} = 0.39488144681077636,$$
$$\bar{\ell}_{\mathrm{SU}(3)} = 0.012860605540213493.$$

Thus

$$\bar{\ell}_{\mathrm{SU}(2)} + \bar{\ell}_{\mathrm{SU}(3)} = 0.40774205235098987,$$
$$P_C/4 = 0.4077420523509898,$$

and

$$\Phi_U(P_C, 0.041124336195630495) = 5.55 \times 10^{-17}.$$

A centered derivative check gives

$$\partial_a \Phi_U(P_C, a) \simeq -10.990524683118785.$$

With $\delta = 10^{-6}$, the bracket signs are

$$\Phi_U(P_C, \alpha_U - \delta) = 1.0990748697592423 \times 10^{-5},$$
$$\Phi_U(P_C, \alpha_U + \delta) = -1.0990300680135956 \times 10^{-5}.$$

For the certified source-only pixel,

$$P_{\mathrm{fwd}} = 1.630972095858897\ldots,$$
$$\alpha_U(P_{\mathrm{fwd}}) = 0.041124247441816685\ldots,$$
$$\frac{v(P_{\mathrm{fwd}})}{E_\star} = 2.0198114078576331 \times 10^{-17}.$$

This branch is the one to cite when avoiding the measured Thomson endpoint in the upstream pixel solve.

**Theorem E.2** (Electroweak unified-coupling numerical witness)**.** *Assume the declared source map, running packet, heat-kernel cutoff policy, and no-hidden-free-variable rule above. Let*

$$I_U = [0.041123336195630494,\ 0.041125336195630496].$$

*If outward-rounded interval arithmetic verifies the directed endpoint signs*

$$\Phi_U(P_C, 0.041123336195630494) > 0, \qquad \Phi_U(P_C, 0.041125336195630496) < 0,$$

*and verifies*

$$0 \notin \partial_a \Phi_U(P_C, I_U),$$

*then there is a unique $\alpha_U(P_C) \in I_U$ satisfying*

$$\Phi_U(P_C, \alpha_U(P_C)) = 0.$$

*The comparison pixel $P_C$ inherits its CODATA electromagnetic input. Holding that pixel fixed, this calculation uses no additional measured low-energy coupling, measured $G$, Planck unit formed using measured $G$, measured $\Lambda$, measured $M_Z$, or measured $M_W$.*

*Proof.* Continuity follows because the running couplings, self-consistent $Z$-scale solution, and finite heat-kernel sums are continuous on the declared positive interval. The endpoint sign change gives existence by the intermediate value theorem. The derivative exclusion gives uniqueness. The dependency statement follows by inspection of the declared source map: the residual is built only from $P_C$, the declared running packet, the self-consistent $Z$-scale equation, and finite compact-group representation sums. □

**Theorem E.3** (Implicit-scale attachment at the comparison pixel)**.** *Let $F(a, \mu)$ be the right-hand side of the source $Z$-scale equation, with the exact stored decimal*

$$\widehat{P}_C = 1.630968209403959324879279847782648941, \qquad J = [7 \times 10^{-18}, 8 \times 10^{-18}].$$

*For every $a \in I_U$, all running denominators are positive on $J$ and*

$$F(a, J) \subset [7.49, 7.52] \times 10^{-18} \subset \operatorname{int} J, \qquad 0 \le \partial_\mu F(a, J) \le 0.006 < 1/100.$$

*There is exactly one scale $\mu(a) \in J$, it is continuously differentiable, and the finite-cutoff hierarchy residual evaluated at this exact scale obeys*

$$\frac{\mathrm{d}}{\mathrm{d}a} \Phi_U(\widehat{P}_C, a) \in [-12, -10].$$

*The lower and upper endpoint values lie respectively in*

$$[1.09907, 1.09908] \times 10^{-5}, \qquad [-1.09904, -1.09902] \times 10^{-5}.$$

*There is therefore exactly one zero in $I_U$. The cutoffs remain $N_2 = 128, N_3 = 64$; neither an infinite representation sum nor a scale outside $J$ is part of this claim.*

*Proof.* The uniform self-map and derivative bound give a contraction on $J$. The interior image and $1 - \partial_\mu F \ge 99/100$ give the implicit derivative

$$\mu'(a) = \frac{\partial_a F}{1 - \partial_\mu F}.$$

Every interval image of a box containing this fixed point retains it. Eight images on $I_U \times J$ enclose the scale for the derivative test; sixteen images at each endpoint enclose the endpoint residuals. These are inclusions of the exact fixed point, not assertions that an iterate equals it. Interval automatic differentiation and an independent analytic calculation verify the displayed bounds at 60-digit interval precision. For the latter, write $S = \alpha_2 + 3\alpha_1/5$ and $B = \alpha_2^2 + 99\alpha_1^2/25$. Then

$$F_\mu = \frac{FB}{4\pi\mu S}, \qquad F_a = F\left[\frac{\pi}{2a^2} + \frac{\alpha_2^2 + 3\alpha_1^2/5}{2a^2 S}\right].$$

The finite entropy derivative is $\mathrm{d}\bar{\ell}_G/\mathrm{d}t = -\operatorname{Cov}_t(C_2, \log d)$; combining it with $t_i' = 4\pi^2\alpha_i^2[a^{-2} + b_i\mu'/(2\pi\mu)]$ gives the certified residual derivative. Continuity, endpoint signs, and strict monotonicity finish the argument. The separate 60- and 64-iteration certificates retain their own finite-map scope and Krawczyk enclosures. □

**Precision policy.** The exponential map makes downstream gravity displays sensitive to $\alpha_U$:

$$\frac{\partial \log G}{\partial \alpha_U} \simeq \frac{\pi}{\alpha_U^2} \approx 1.86 \times 10^3.$$

Therefore the number of digits printed for any downstream $G$ or $\varepsilon_{\mathrm{Cs}}$ row may not exceed the interval precision certified by

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}_{\mathrm{QCD/nuc}}^{133\mathrm{Cs}} + \mathcal{R}_{\mathrm{atom}}^{133\mathrm{Cs}}.$$

The witness supports an electroweak hierarchy certificate. It does not support a 52-digit source-only gravity prediction.

## E.2 One-Loop Calibration Frame

At one loop, if couplings unify at $(M_U, \alpha_U)$,

$$\alpha_i^{-1}(M_Z) = \alpha_U^{-1} + \frac{b_i}{2\pi}\ln\frac{M_U}{M_Z}.$$

On the forward calibration branch these are the values $(M_U(P), \alpha_U(P))$ emitted by the source solve. They are not inferred by inverse readback from measured low-energy couplings. Writing $A_i := \alpha_i^{-1}(M_Z)$ and $L := \ln(M_U/M_Z)$, one finds

$$L = \frac{2\pi}{b_1 - b_2}(A_1 - A_2),$$

and the corresponding consistency relation for this calibration is

$$A_3^{\mathrm{pred}} = \frac{b_3 - b_2}{b_1 - b_2}A_1 + \frac{b_1 - b_3}{b_1 - b_2}A_2.$$

This appendix keeps that algebra on the page as a calibration relation, not as a theorem that OPH derives a supersymmetric UV spectrum.

### E.3 Benchmark coefficient comparison

For the standard one-loop benchmark coefficients,

| Model | $(b_1, b_2, b_3)$ | $\alpha_s(M_Z)^{\mathrm{pred}}$ |
|---|---|---|
| Standard Model | $(41/10, -19/6, -7)$ | $\approx 0.071$ |
| minimal supersymmetric Standard Model | $(33/5, 1, -3)$ | $\approx 0.116$ |
| Observed [24] | | $\approx 0.118$ |

At one loop, the printed minimal supersymmetric Standard Model (MSSM) coefficients reproduce the observed closure far better than the plain Standard Model coefficients. The forward calibration matches this benchmark. It does not imply that OPH derives a supersymmetric ultraviolet spectrum. The observed row is the Particle Data Group world average, quoted at the precision this benchmark separates, and the one-loop comparison is the textbook unification benchmark rather than an output of the OPH source, which is why it carries no claim class and no comparison row of its own.

### E.4 Heat-Kernel Sector Law

Read this subsection as the compact-group merge boundary above the fixed-cutoff microphysics theorem. MaxEnt yields the familiar $d_R e^{-tC_2(R)}$ law only after the quadratic-Casimir generator and one-sided edge algebra below are declared; the Peter–Weyl lift alone does not select either input. This is not a second, independent proof of the finite-cutoff same-overlap Casimir branch.

**Theorem E.4** (Heat-kernel edge-sector weights)**.** *Let the MaxEnt edge generator be the factorwise group Laplacian $H_{\mathrm{edge}} = \sum_i c_i \Delta_{G_i}$, with eigenvalue $C_2(R)$ in the one-factor notation, and let the probability space be the one-sided electric-center algebra $\bigoplus_R B(V_R)$, where the sector projector has trace $d_R$. Then the sector probabilities take heat-kernel form:*

$$p_R(t) \propto d_R\, e^{-tC_2(R)},$$

*where $d_R = \dim R$ and $C_2(R)$ is the quadratic Casimir. Bi-invariance by itself permits other central energies, such as $C_2(R)^2$. A Gibbs trace on the two-sided Peter–Weyl space $L^2(G)$ instead gives the multiplicity $d_R^2$.*

**Lemma E.5** (Bi-invariant operators)**.** *If $G = \prod_i G_i$ is a compact semisimple Lie group written as a product of compact simple factors, then any bi-invariant second-order differential operator on $G$ has the form*

$$D = c_0 \mathbf{1} - \sum_i c_i \Delta_{G_i},$$

*where $\Delta_{G_i}$ is the Laplace–Beltrami operator on the factor $G_i$.*

**Remark E.6.** *Bi-invariance is equivalent to $D \in Z(U(\mathfrak{g}))$. Linear terms vanish because there are no invariant vectors in the adjoint representation, and the quadratic part is proportional to the Killing form on each simple factor, giving one independent coefficient per factor. The Casimir element then acts as $-\Delta_G$ in the regular representation.*

### E.5 Peter–Weyl Multiplicity and Beta Shifts

By Peter–Weyl decomposition, the effective refinement-limit edge representation space is

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*.$$

Entanglement traces over one factor give multiplicity $d_R$ in $p_R$, but loops see both factors, so the effective multiplicity entering the calibration branch is

$$N_{\text{eff}}(R) = d_R\, p_R.$$

The one-loop beta shift from edge sectors is then

$$\Delta b_a = \sum_R p_R\, d_R\, T_a(R),$$

where $T_a(R)$ is the Dynkin index for gauge factor $a$. This is the exact mathematical point where edge-sector heat-kernel weights feed the calibration branch. Matching to MSSM-like one-loop running is a comparison benchmark, not a theorem-level MSSM spectrum claim.

The branch boundary for theorem-level unification closure is explicit. One requires a derived refinement-limit identification of which transportable sectors actually contribute to the running carrier, a derived statistics/grading rule rather than an imposed fermionic-grading restriction, controlled threshold and decoupling conventions on that same carrier, a derived representation content/truncation rule for the sectors retained in the comparison, and a proof that the effective loop multiplicities entering $\Delta b_a$ are exactly the ones used in this calibration package rather than nearby alternatives. The declared runtime surface does not emit the full RG/matching/threshold/scheme packet: scheme lock, threshold map, beta provenance, and interval composition are separate certificate requirements.

At the branch unification diffusion parameter $t_U(P) \approx 1.64$ selected by the forward source solve, the benchmark shift is

$$\Delta b \approx (2.49, 4.38, 3.97) \qquad \text{vs MSSM} \qquad (2.50, 4.17, 4.00).$$

With the additional calibration assumption of a fermionic-grading restriction to half-integer SU(2) sectors, this sharpens to

$$\Delta b \approx (2.50, 4.17, 3.97),$$

which matches the MSSM-style one-loop benchmark at the percent level under that declared calibration package. Equivalently, the benchmark ratio

$$\frac{\Delta b_3}{\Delta b_2} \approx 0.91$$

sits close to the MSSM comparison value 0.96.

## F H3 Worldline-Stitch Certificate

This appendix records the finite evidence-facing certificate for Theorem D.5. It is separate from the heat-kernel calibration package above. A passing certificate is a record-continuation statement in a declared $H^3$ observer chart, not a particle-species, mass, charge, or scattering-amplitude theorem.

The certificate payload has the following required blocks.

1. **Atlas.** It declares the hyperboloid model $H^3_{R_H} \subset \mathbb{R}^{1,3}$, the Lorentz signature $(-+++)$, the curvature radius $R_H$, chart domains, and chart transitions $G_{ji} \in \mathrm{SO}^+(1,3)$. Transition residuals must certify Lorentz inner-product preservation, orientation, and future-sheet preservation.

2. **Clock map.** Each local record interval is mapped to a common comparison-time line. The observer-time adjacency margin must dominate the declared clock uncertainty.

3. **Chart-natural extraction.** Raw local records are connected components of a frozen detector scalar on the $H^3$ chart. The detector scalar, thresholds, support collars, and chart-naturality residuals are included. Implementation record IDs are not inputs.

4. **Overlap descent.** Records are first descended across genuine chart overlaps. The payload records same-component join margins, distinct-component separation margins, support errors, and triple-overlap cocycle residuals.

5. **Interface crossing.** A candidate stitch must cross a real interface in adjacent clock intervals. It includes oriented signed-distance margins and a positive lower bound on normal velocity. Tangential, grazing, and file-boundary-only contacts are rejected.

6. **Transport.** The candidate edge is evaluated in a common chart after sector and gauge transport. The atlas holonomy comparison is separate from the gauge-holonomy comparison.

7. **Assignment.** The matching is one-to-one with appearance/disappearance interval costs. The proposed winner has an upper cost bound, every competitor has a lower cost bound, and

$$\Delta_{\mathrm{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*) > 0.$$

   Record IDs, global IDs, stitch keys, and shard-local IDs are forbidden in admissibility, costs, and tie-breaking.

8. **Refinement.** A coarse/fine pair supplies $Q_{sr}$, a distortion bound $\eta_{sr}$, and a contracted-graph isomorphism check. The coarse gap must satisfy $\Delta_r > 2\eta_{sr}$.

9. **Interaction firewall.** A free-propagation slab is required. If an interaction marker is present, the terminal label is H3 interaction required and the event is handed to the interaction solver.

The allowed terminal outcomes are

H3 stitch certified, H3 stitch ambiguous, H3 stitch rejected,
H3 interaction required, H3 atlas invalid, H3 certificate incomplete.

H3 stitch certified is emitted only when every block above passes and the stable assignment gap is positive. Equal-cost matchings, preserved nontrivial permutations of the visible data, or coarse/fine gaps below margin make the fail-closed protocol emit H3 stitch ambiguous, meaning that it has certified no unique stitch. This verdict is not by itself a theorem of mathematical nonuniqueness; a symmetry obstruction requires a proved action that is fixed-point-free on the admissible stitch set. Missing fields force H3 certificate incomplete. Failed atlas, interface, metric, or transport gates force rejection or atlas invalidity. In particular, Euclidean distances on the stored coordinate triples are not certificate distances; the certificate distance is the hyperboloid geodesic $d_H$ from Section D.

## Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- A5 flavor potentials and residual-stabilizer selection [56, 74].
- Burnside_orbit_counting [58, 48].
- Classical electroweak symmetry breaking and Goldstone counting in the Standard Model Higgs sector [110, 97, 65].
- Classical Koide relation, standard one-loop running and the existing OPH certified alpha and tau artifacts [75, 14, 97].
- complex_pole_to_running_width_Breit_Wigner_conversion [43, 103].
- cumulant expansions [76].
- De Broglie internal-clock and Zitterbewegung phase kinematics [54, 67, 101].
- Electroweak phenomenology, exterior weak multiplicities, hierarchy-problem literature, and commuting-square naturality diagnostics [22, 42, 105, 106, 82].
- exact_quadratic_field_linear_algebra [50].
- exact_rational_linear_algebra [107, 44].
- Exterior or Fock algebra and Standard Model exterior embeddings [42, 47, 61].
- exterior_algebra_and_CAR [47, 71].
- Finite clock models, Hamiltonian scaling, and determinant-line mass invariants [96, 68].
- Finite-group heat kernels [55, 100].
- finite_absorbing_Markov_chains_and_triangular_spectra [72, 68].
- finite_group_invariant_operator_averaging [102].
- finite_group_representation_theory_and_Schur_lemma [102, 60].
- finite_irreducible_periodic_Markov_chain_stationarity [77, 92].
- finite_random_unitary_channels_and_class_generators [109, 55].
- Flavor-basis invariants, singular-value decompositions, and parameter non-identifiability [70, 45, 98].
- Free plane-wave evolution and de Broglie phase kinematics under translation flows [54, 111].
- Georgi–Jarlskog/Clebsch flavor textures, one-loop Yukawa running, and lattice light-quark ratios [62, 81, 20].
- Harremoes and Topsoe (2001), DOI 10.1109/18.959272, Shannon entropy at fixed index of coincidence [18].
- Koide's charged-lepton relation, Hermitian C3 circulants, and finite GNS constructions [75, 52, 46].
- maximal_matching_combinatorics [78].
- moment_and_cumulant_algebra [89, 95].
- numerical_quadrature_and_coupled_ODE_shooting [104, 53].

- Pauli algebra sigma_a sigma_b = delta_ab + i epsilon_abc sigma_c and the Pauli Hamiltonian derivation of g = 2 [94, 99].
- PDG and FLAG quark summaries [22, 20].
- perturbative renormalization-group transport [80, 79].
- positive_Hermitian_kinetic_normalization [68, 73].
- Primary mathematical and physical sources [42, 91].
- QCD running-to-pole conversion [49, 90].
- Residual-symmetry flavor mixing and icosahedral axis geometry [74, 56].
- scalar_ODE_parameter_sensitivity_and_comparison_barriers [66, 108].
- Schur_complement_differentiation [113, 83].
- Signed graph Laplacians, integer positive-definite matrices, and spectral residual estimates [112, 69, 93].
- SMDR_1_3_complex_pole_and_local_charged_current_matching [88, 87].
- Standard Model Higgs near-criticality calculations [59, 23].
- Standard_Model_beta_functions_top_only_specialization [81, 79].
- Standard_Model_two_loop_beta_functions_and_effective_potential [79, 57, 84].
- standard_SU3_fundamental_Casimir [63, 97].
- strict_convexity_and_exponential_family_KL_projection [51, 41].
- SU3_and_SU2_intertwiner_identities [63, 60].
- Tree-level charged-current couplings g/sqrt(2) with weak-isospin raising and lowering operators [110, 97].
- Tree-level electroweak neutral-current couplings Q = T3 + Y and T3 - Q sin^2 theta [64, 97, 110].
- TSIL_and_3VIL_loop_integral_evaluation [85, 86].

## References


[1] B. Müller et al., *Observer-Patch Holography: formal proofs and executable receipts*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2054/Lean/Screen

[2] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026.
Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/extra/observable_normal_forms.pdf.

[3] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, J. Kim, D. Matscheko, D. A. Friedman, J. Hill, W. T. Glynn, M. A. Visser, K. A. Anirudha, and B. de La Fournière, *Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/flagship/from_observer_consensus_to_standard_physics.pdf

[4] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, and D. A. Friedman, *Observers Are All You Need*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/paper/observers_are_all_you_need.pdf

[5] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/paper/reality_as_consensus_protocol.pdf

[6] B. Müller, A. Osika, K. Xue, B. Cassie, B. de La Fournière, and D. A. Friedman, *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in Observer-Patch Holography*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/paper/screen_microphysics_and_observer_synchronization.pdf

[7] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Source: https://github.com/muellerberndt/oph-physics-sim. Canonical evidence archives: https://github.com/FloatingPragma/observer-patch-holography/tree/r2054/evidence. Exact local-domain archive and checker: https://github.com/FloatingPragma/observer-patch-holography/tree/r2054/evidence/local_domain.

[8] FloatingPragma, *Particle-simulation receipt archive*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2054/evidence/particle_simulation_receipts.

[9] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[10] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[11] B. Müller, M. Poneder, J. Kim, and D. A. Friedman, *An Executable Observer for Observer Patch Holography: Specification, Certified Components and Acceptance Criteria*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/paper/executable_observer_construction.pdf

[12] FloatingPragma, *Repair laws: what is fixed, what is free and how to tell*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/model/CANDIDATES.md

[13] National Institute of Standards and Technology, *2022 CODATA Recommended Values of the Fundamental Constants of Physics and Chemistry*, NIST SP 959, May 2024. https://physics.nist.gov/cuu/pdf/wallet_2022.pdf

[14] Y. Koide, *New view of quark and lepton mass hierarchy*, Physical Review D **28**, 252 (1983).

[15] E. Layton, Y. Huang, and S.-I. Chu, *Cyclic quantum evolution and Aharonov–Anandan geometric phases in SU(2) spin-coherent states*, Physical Review A **41**, 42 (1990). https://doi.org/10.1103/PhysRevA.41.42

[16] E. M. Rabei, Arvind, R. Simon, and N. Mukunda, *Bargmann invariants and geometric phases—a generalized connection*, Physical Review A **60**, 3397 (1999). https://arxiv.org/abs/quant-ph/9907080

[17] C. A. Brannen, *The Lepton Masses*, preprint, 2006. http://brannenworks.com/MASSES2.pdf

[18] P. Harremoës and F. Topsøe, *Inequalities between entropy and index of coincidence derived from information diagrams*, IEEE Transactions on Information Theory **47**, 2944–2960 (2001). https://doi.org/10.1109/18.959272

[19] S. Antusch, K. Hinze, and S. Saad, *Updated Running Quark and Lepton Parameters at Various Scales*, arXiv:2510.01312, 2025. https://arxiv.org/abs/2510.01312

[20] Y. Aoki et al. (Flavour Lattice Averaging Group), *FLAG Review 2024*, arXiv:2411.04268, 2024, Tables 11–12. https://arxiv.org/abs/2411.04268

[21] S. Navas et al. (Particle Data Group), *CKM Quark-Mixing Matrix*, Review of Particle Physics 2024, Sec. 12.2.2, revised May 31, 2024. https://pdg.lbl.gov/2024/reviews/rpp2024-rev-ckm-matrix.pdf

[22] S. Navas et al. (Particle Data Group), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024), 2025 update, quark summary table. https://pdg.lbl.gov/2025/tables/rpp2025-sum-quarks.pdf

[23] D. Buttazzo et al., *Investigating the near-criticality of the Higgs boson*, JHEP **12** (2013) 089, Appendix B. https://arxiv.org/abs/1307.3536

[24] Particle Data Group, *Review of Particle Physics: 2026 particle listings*, 2026. https://pdg.lbl.gov/2026/listings/particle_properties.html

[25] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astronomy & Astrophysics 641, A6, 2020. https://arxiv.org/abs/1807.06209

[26] DESI Collaboration, *Constraints on Neutrino Physics from DESI DR2 BAO and DR1 Full Shape*, arXiv:2503.14744, 2025. https://arxiv.org/abs/2503.14744

[27] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, JHEP **12** (2024) 216, arXiv:2410.05380, with the NuFIT 6.1 (2025) profile-table release at https://www.nu-fit.org/?q=node/309.

[28] B. A. Kniehl, *All-order renormalization of the propagator matrix for fermionic systems with flavor mixing*, Physical Review D **89**, 096005 (2014), arXiv:1308.3140.

[29] P. Gambino and P. A. Grassi, *The Nielsen identities of the Standard Model and the definition of mass*, Physical Review D **62**, 076002 (2000), arXiv:hep-ph/9907254.

[30] D. Buchholz, *Gauss' law and the infraparticle problem*, Physics Letters B **174**, 331–334 (1986).

[31] P. Duch and W. Dybalski, *Infrared problem in quantum electrodynamics*, arXiv:2307.06114.

[32] M. Lüscher, *Abelian chiral gauge theories on the lattice with exact gauge invariance*, Nuclear Physics B **549**, 295–334 (1999), arXiv:hep-lat/9811032.

[33] M. Lüscher, *Lattice regularization of chiral gauge theories to all orders of perturbation theory*, Journal of High Energy Physics **06** (2000) 028, arXiv:hep-lat/0006014.

[34] D. Kadoh and Y. Kikukawa, *A simple construction of fermion measure term in $U(1)$ chiral lattice gauge theories with exact gauge invariance*, Journal of High Energy Physics **02** (2008) 063, arXiv:0709.3658.

[35] H. Bélusca-Maïto, A. Ilakovac, M. Mađor-Božinović, and D. Stöckinger, *Dimensional regularization and Breitenlohner–Maison/'t Hooft–Veltman scheme for $\gamma_5$ applied to chiral Yang–Mills theory*, Journal of High Energy Physics **08** (2020) 024, arXiv:2004.14398.

[36] V. Dūdėnas and M. Löschner, *Vacuum expectation value renormalization in the Standard Model and beyond*, Physical Review D **103**, 013003 (2021), arXiv:2010.15076.

[37] S. Dittmaier and H. Rzehak, *Electroweak renormalization based on gauge-invariant vacuum expectation values of non-linear Higgs representations*, Journal of High Energy Physics **05** (2022) 125, arXiv:2203.07236.

[38] P. A. Grassi, B. A. Kniehl, and A. Sirlin, *Width and partial widths of unstable particles in the light of the Nielsen identities*, Physical Review D **65**, 085001 (2002), arXiv:hep-ph/0109228.

[39] S. P. Martin and D. G. Robertson, *Standard Model parameters in the tadpole-free pure $\overline{\text{MS}}$ scheme*, arXiv:1907.02500.

[40] S. P. Martin, *Top-quark pole mass in the tadpole-free MS-bar scheme*, arXiv:1604.01134.

[41] Amari, S. i., and Nagaoka, H. (2000). *Methods of Information Geometry*. American Mathematical Society. https://doi.org/10.1090/mmono/191

[42] Baez, J., and Huerta, J. (2010). "The algebra of grand unified theories." *Bulletin of the American Mathematical Society* **47**, 483–552. https://doi.org/10.1090/S0273-0979-10-01294-2

[43] Bardin, D. Y., Leike, A., Riemann, T., and Sachwitz, M. (1988). "Energy-dependent width effects in e+e- annihilation near the Z-boson pole." *Physics Letters B* **206**, 539–542. https://doi.org/10.1016/0370-2693(88)91627-9

[44] Bareiss, E. H. (1968). "Sylvester's identity and multistep integer-preserving Gaussian elimination." *Mathematics of Computation* **22**, 565–578. https://doi.org/10.1090/S0025-5718-1968-0226829-0

[45] Branco, G. C., Lavoura, L., and Silva, J. P. (1999). *CP Violation*. Clarendon Press, Oxford. ISBN 0-19-850399-7.

[46] Bratteli, O., and Robinson, D. W. (1987). *Operator Algebras and Quantum Statistical Mechanics 1: C*- and W*-Algebras, Symmetry Groups, Decomposition of States*. Springer. https://doi.org/10.1007/978-3-662-02520-8

[47] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2*. Springer. https://doi.org/10.1007/978-3-662-03444-6

[48] Burnside, W. (1911). *Theory of Groups of Finite Order*. Cambridge University Press. https://doi.org/10.1017/CBO9781139237253

[49] Chetyrkin, K., and Steinhauser, M. (2000). "The relation between the $\overline{\text{MS}}$ and the on-shell quark mass at order $\alpha_s^3$." *Nuclear Physics B* **573**, 617–651. https://doi.org/10.1016/s0550-3213(99)00784-1

[50] Cohen, H. (1993). *A Course in Computational Algebraic Number Theory*. Springer. https://doi.org/10.1007/978-3-662-02945-9

[51] Csiszár, I. (1975). "I-divergence geometry of probability distributions and minimization problems." *Annals of Probability* **3**, 146–158. https://doi.org/10.1214/aop/1176996454

[52] Davis, P. J. (1979). *Circulant Matrices*. Wiley, New York. ISBN 0-471-05771-1.

[53] Davis, P. J., and Rabinowitz, P. (1984). *Methods of Numerical Integration*. Academic Press. https://doi.org/10.1016/C2013-0-10566-1

[54] de Broglie, L. (1925). "Recherches sur la théorie des quanta." *Annales de Physique* **10**, 22–128. https://doi.org/10.1051/anphys/192510030022

[55] Diaconis, P. (1988). *Group Representations in Probability and Statistics*. Institute of Mathematical Statistics (Lecture Notes–Monograph Series). https://doi.org/10.1214/lnms/1215467407

[56] Everett, L. L., and Stuart, A. J. (2009). "Icosahedral (A5) family symmetry and the golden ratio prediction for solar neutrino mixing." *Physical Review D* **79**, 085005. https://doi.org/10.1103/PhysRevD.79.085005

[57] Ford, C., Jack, I., and Jones, D. (1992). "The standard model effective potential at two loops." *Nuclear Physics B* **387**, 373–390. https://doi.org/10.1016/0550-3213(92)90165-8

[58] Frobenius, G. (1887). "Ueber die Congruenz nach einem aus zwei endlichen Gruppen gebildeten Doppelmodul." *Journal für die reine und angewandte Mathematik* **101**, 273–299. https://doi.org/10.1515/crll.1887.101.273

[59] Froggatt, C., and Nielsen, H. (1996). "Standard Model criticality prediction: top mass 173 ± 5 GeV and Higgs mass 135 ± 9 GeV." *Physics Letters B* **368**, 96–102. https://doi.org/10.1016/0370-2693(95)01480-2

[60] Fulton, W., and Harris, J. (1991). *Representation Theory: A First Course*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4612-0979-9

[61] Georgi, H., and Glashow, S. L. (1974). "Unity of All Elementary-Particle Forces." *Physical Review Letters* **32**, 438–441. https://doi.org/10.1103/PhysRevLett.32.438

[62] Georgi, H., and Jarlskog, C. (1979). "A new lepton-quark mass relation in a unified theory." *Physics Letters B* **86**, 297–300. https://doi.org/10.1016/0370-2693(79)90842-6

[63] Georgi, H. (1999). *Lie Algebras in Particle Physics: From Isospin to Unified Theories*. Perseus Books. ISBN 0-7382-0233-9.

[64] Glashow, S. L. (1961). "Partial-symmetries of weak interactions." *Nuclear Physics* **22**, 579–588. https://doi.org/10.1016/0029-5582(61)90469-2

[65] Goldstone, J., Salam, A., and Weinberg, S. (1962). "Broken Symmetries." *Physical Review* **127**, 965–970. https://doi.org/10.1103/PhysRev.127.965

[66] Hartman, P. (2002). *Ordinary Differential Equations*. Society for Industrial and Applied Mathematics (Classics in Applied Mathematics). https://doi.org/10.1137/1.9780898719222

[67] Hestenes, D. (1990). "The zitterbewegung interpretation of quantum mechanics." *Foundations of Physics* **20**, 1213–1232. https://doi.org/10.1007/BF01889466

[68] Horn, R. A., and Johnson, C. R. (2012). *Matrix Analysis*. Cambridge University Press. https://doi.org/10.1017/CBO9781139020411

[69] Hou, Y., Li, J., and Pan, Y. (2003). "On the Laplacian Eigenvalues of Signed Graphs." *Linear and Multilinear Algebra* **51**, 21–30. https://doi.org/10.1080/0308108031000053611

[70] Jarlskog, C. (1985). "Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation." *Physical Review Letters* **55**, 1039–1042. https://doi.org/10.1103/PhysRevLett.55.1039

[71] Jordan, P., and Wigner, E. (1928). "Über das Paulische Äquivalenzverbot." *Zeitschrift für Physik* **47**, 631–651. https://doi.org/10.1007/BF01331938

[72] Kemeny, J. G., and Snell, J. L. (1976). *Finite Markov Chains*. Springer-Verlag, New York (Undergraduate Texts in Mathematics). ISBN 0-387-90192-2.

[73] King, S., and Peddie, I. (2004). "Canonical normalisation and Yukawa matrices." *Physics Letters B* **586**, 83–94. https://doi.org/10.1016/j.physletb.2004.01.087

[74] King, S. F., and Luhn, C. (2013). "Neutrino mass and mixing with discrete symmetry." *Reports on Progress in Physics* **76**, 056201. https://doi.org/10.1088/0034-4885/76/5/056201

[75] Koide, Y. (1983). "A fermion-boson composite model of quarks and leptons." *Physics Letters B* **120**, 161–165. https://doi.org/10.1016/0370-2693(83)90644-5

[76] Kubo, R. (1962). "Generalized cumulant expansion method." *Journal of the Physical Society of Japan* **17**, 1100–1120. https://doi.org/10.1143/JPSJ.17.1100

[77] Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*. American Mathematical Society. https://doi.org/10.1090/mbk/107

[78] Lovász, L., and Plummer, M. D. (2009). *Matching Theory*. AMS Chelsea Publishing, Providence. https://doi.org/10.1090/chel/367

[79] Luo, M., Wang, H., and Xiao, Y. (2003). "Two-loop renormalization group equations in general gauge field theories." *Physical Review D* **67**, 065019. https://doi.org/10.1103/PhysRevD.67.065019

[80] Machacek, M. E., and Vaughn, M. T. (1983). "Two-loop renormalization group equations in a general quantum field theory." *Nuclear Physics B* **222**, 83–103. https://doi.org/10.1016/0550-3213(83)90610-7

[81] Machacek, M. E., and Vaughn, M. T. (1984). "Two-loop renormalization group equations in a general quantum field theory (II). Yukawa couplings." *Nuclear Physics B* **236**, 221–232. https://doi.org/10.1016/0550-3213(84)90533-9

[82] Mac Lane, S. (1998). *Categories for the Working Mathematician*. Springer. https://doi.org/10.1007/978-1-4757-4721-8

[83] Magnus, J. R., and Neudecker, H. (2019). *Matrix Differential Calculus with Applications in Statistics and Econometrics*. Wiley. https://doi.org/10.1002/9781119541219

[84] Martin, S. P. (2002). "Two-loop effective potential for a general renormalizable theory and softly broken supersymmetry." *Physical Review D* **65**, 116003. https://doi.org/10.1103/PhysRevD.65.116003

[85] Martin, S. P., and Robertson, D. G. (2006). "TSIL: a program for the calculation of two-loop self-energy integrals." *Computer Physics Communications* **174**, 133–151. https://doi.org/10.1016/j.cpc.2005.08.005

[86] Martin, S. P., and Robertson, D. G. (2017). "Evaluation of the general three-loop vacuum Feynman integral." *Physical Review D* **95**, 016008. https://doi.org/10.1103/PhysRevD.95.016008

[87] Martin, S. P. (2019). "Matching relations for decoupling in the standard model at two loops and beyond." *Physical Review D* **99**, 033007. https://doi.org/10.1103/PhysRevD.99.033007

[88] Martin, S. P., and Robertson, D. G. (2019). "Standard Model parameters in the tadpole-free pure MS-bar scheme." *Physical Review D* **100**, 073004. https://doi.org/10.1103/PhysRevD.100.073004

[89] McCullagh, P. (1987). *Tensor Methods in Statistics*. Chapman and Hall, London. ISBN 0-412-27480-9.

[90] Melnikov, K., and van Ritbergen, T. (2000). "The three-loop relation between the MS-bar and the pole quark masses." *Physics Letters B* **482**, 99–108. https://doi.org/10.1016/S0370-2693(00)00507-4

[91] Nielsen, M. A., and Chuang, I. L. (2010). *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press. https://doi.org/10.1017/CBO9780511976667

[92] Norris, J. R. (1997). *Markov Chains*. Cambridge University Press. https://doi.org/10.1017/CBO9780511810633

[93] Parlett, B. N. (1998). *The Symmetric Eigenvalue Problem*. Society for Industrial and Applied Mathematics. https://doi.org/10.1137/1.9781611971163

[94] Pauli, W. (1927). "Zur Quantenmechanik des magnetischen Elektrons." *Zeitschrift für Physik* **43**, 601–623. https://doi.org/10.1007/BF01397326

[95] Peccati, G., and Taqqu, M. S. (2011). *Wiener Chaos: Moments, Cumulants and Diagrams*. Springer, Milan (Bocconi & Springer Series). https://doi.org/10.1007/978-88-470-1679-8

[96] Peres, A. (1980). "Measurement of time by quantum clocks." *American Journal of Physics* **48**, 552–557. https://doi.org/10.1119/1.12061

[97] Peskin, M. E., and Schroeder, D. V. (1995). *An Introduction to Quantum Field Theory*. Addison-Wesley. https://doi.org/10.1201/9780429503559

[98] Rothenberg, T. J. (1971). "Identification in Parametric Models." *Econometrica* **39**, 577–591. https://doi.org/10.2307/1913267

[99] Sakurai, J. J., and Napolitano, J. (2020). *Modern Quantum Mechanics*. Cambridge University Press. https://doi.org/10.1017/9781108587280

[100] Saloff-Coste, L. (2004). "Random Walks on Finite Groups." In *Probability on Discrete Structures*, 263–346. Springer, Berlin (Encyclopaedia of Mathematical Sciences). https://doi.org/10.1007/978-3-662-09444-0_5

[101] Schrödinger, E. (1930). "Über die kräftefreie Bewegung in der relativistischen Quantenmechanik." *Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, 418–428. https://zbmath.org/?q=an:56.0754.06

[102] Serre, J. P. (1977). *Linear Representations of Finite Groups*. Springer. https://doi.org/10.1007/978-1-4684-9458-7

[103] Sirlin, A. (1991). "Theoretical considerations concerning the Z0 mass." *Physical Review Letters* **67**, 2127–2130. https://doi.org/10.1103/PhysRevLett.67.2127

[104] Stoer, J., and Bulirsch, R. (2002). *Introduction to Numerical Analysis*. Springer, New York (Texts in Applied Mathematics). https://doi.org/10.1007/978-0-387-21738-3

[105] Susskind, L. (1979). "Dynamics of spontaneous symmetry breaking in the Weinberg-Salam theory." *Physical Review D* **20**, 2619–2625. https://doi.org/10.1103/PhysRevD.20.2619

[106] 't Hooft, G. (1980). "Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking." In *Recent Developments in Gauge Theories*, 135–157. Springer. https://doi.org/10.1007/978-1-4684-7571-5_9

[107] von zur Gathen, J., and Gerhard, J. (2013). *Modern Computer Algebra*. Cambridge University Press. https://doi.org/10.1017/CBO9781139856065

[108] Walter, W. (1970). *Differential and Integral Inequalities*. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-86405-6

[109] Watrous, J. (2018). *The Theory of Quantum Information*. Cambridge University Press. https://doi.org/10.1017/9781316848142

[110] Weinberg, S. (1967). "A model of leptons." *Physical Review Letters* **19**, 1264–1266. https://doi.org/10.1103/PhysRevLett.19.1264

[111] Weinberg, S. (1995). *The Quantum Theory of Fields, Volume 1: Foundations*. Cambridge University Press. https://doi.org/10.1017/CBO9781139644167

[112] Zaslavsky, T. (1982). "Signed graphs." *Discrete Applied Mathematics* **4**, 47–74. https://doi.org/10.1016/0166-218X(82)90033-6

[113] Zhang, F. (2005). *The Schur Complement and Its Applications*. Springer. https://doi.org/10.1007/b105056

[114] D. Burgarth, P. Facchi, H. Nakazato, S. Pascazio and K. Yuasa, "Quantum Zeno Dynamics from General Quantum Operations," *Quantum* **4**, 289 (2020), arXiv:1809.09570.

[115] L. Susskind, "Lattice fermions," *Phys. Rev. D* **16**, 3031–3039 (1977).

[116] T. Misumi, T. Onogi and T. Yamaoka, "Taste-splitting mass and edge modes in 3+1D staggered fermions," arXiv:2604.02078v1 (2026), Sections 2.1–2.2.

[117] P. Caputo, T. M. Liggett and T. Richthammer, "Proof of Aldous' spectral gap conjecture," *J. Amer. Math. Soc.* **23**, 831–851 (2010), arXiv:0906.1238, Theorem 1.1 and Section 4.1.1.

## AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.